Sample Description:

Sample quality is optimum for the test.

Variants of Uncertain Significance identified in relation to phenotype

Key Findings:

Gene & Transcript	Location	Variant	Zygosity/	Phenotype	Clinical
KIF26A (+) NM_015656.2	Intron 6	c.1326+2T>A (Splice donor variant)	Likely Compound Heterozygous /Autosomal Recessive	Cortical dysplasia, complex, with other brain malformations 11	Uncertain significance# (PM2, PVS1)
KIF26A (+) NM_015656.2	Exon 3	c.371G>A (p.Arg124His)			Uncertain significance# (PM2)

**Genetic test results are reported based on the recommendations of American College of Medical Genetics*

 # Due to lack of clinical evidence for this variant, it is classified as variant of uncertain significance.

Note: The variant KIF26A:c.1326+2T>A to be confirmed by sanger due to low death of the variant.

Variant Interpretation &Clinical correlation

***KIF26A*:c.1326+2T>A;**

The submitted sample shows Heterozygous variant in Intron 6 of gene *KIF26A* (chr14: g.104171937T>A). A splice donor variant “T” to “A” detected at nucleotide position 1326+2. This variant mutates a splice-donor sequence, potentially resulting in the retention of large segments of intronic DNA by the mRNA and nonfunctional proteins. ***In silico predictions:*** The c.1326+2T>A variant is a loss of function variant in the gene KIF26A, which is intolerant of Loss of Function variant. ***Population frequency and Internal database***: The c.1326+2T>A variant is novel (not in any individuals) in gnomAD All. The c.1326+2T>A variant is novel (not in any individuals) in 1kG All. This variant is absent in our internal database.

Based on above evidence, this variant (KIF26A:c.1326+2T>A) is classified as* Uncertain significance *variant.

***KIF26A*:c.371G>A;**

The submitted sample shows Heterozygous variant in Exon 3 of gene *KIF26A* (chr14: g.104152097G>A). A missense variant "G" to "A" detected at nucleotide position 371 leading to change in amino acid sequences from Arginine to Histidine at codon 124. ***In silico predictions:*** The p.Arg124His missense variant is predicted to be tolerated by both SIFT or PolyPhen2. The nucleotide c.371 in KIF26A is not conserved according to a GERP++ and PhyloP analysis of 100 vertebrates. ***Population frequency and Internal database*:** The p.Arg124His variant is observed in 3/30,692 (0.0098%) alleles from individuals of gnomAD South Asian background in gnomAD All. The p.Arg124His variant is novel (not in any individuals) in 1kG All. This variant is absent in our internal database.

Based on above evidence, this variant (KIF26A:c.371G>A) is classified as* Uncertain significance *variant.

***KIF26A Gene Phenotype Description*:**
An autosomal recessive disorder of aberrant neuronal migration during brain development. CDCBM11 is characterized by dilated ventricles and reduced white matter, and is associated with axonal developmental defects.

Additional Findings:

Gene & Transcript	Location	Variant &	Zygosity/	Phenotype	Clinical
DMD (-) NM_004006.3	Xp21.1 (?_31,773,949- 31,875,375_?) Exon 48-51	Deletion c.6913-?_c.7542+? 101.4kb	Hemizygous /X-linked Recessive	Duchenne muscular dystrophy	Pathogenic

**Genetic test results are reported based on the recommendations of American College of Medical Genetics*

***DMD Gene Phenotype Description*:**
Most common form of muscular dystrophy; a sex-linked recessive disorder. It typically presents in boys aged 3 to 7 year as proximal muscle weakness causing waddling gait, toe-walking, lordosis, frequent falls, and difficulty in standing up and climbing up stairs. The pelvic girdle is affected first, then the shoulder girdle. Progression is steady and most patients are confined to a wheelchair by age of 10 or 12. Flexion contractures and scoliosis ultimately occur. About 50% of patients have a lower IQ than their genetic expectations would suggest. There is no treatment.

- Genetic counseling is advised for interpretation on the consequences of the variant(s).
- We recommend confirming the presence of these variants by Sanger Sequencing/ MLPA.
- Segregation analysis of the variant by Sanger /Exome sequencing/ MLPA is recommended.
- The significance/classification of the variants might change based on parental and family members genetic testing.
- For questions about this report, or for assistance in locating nearby genetic counseling services, please contact the Laboratory.
- If results obtained do not match the clinical findings, additional testing should be considered as per referring clinician's recommendation.

Methodology:

- Next Generation Sequencing: DNA extracted from Blood, Saliva, Amniotic fluid, CVS or any other standard source is used for targeted capture-based Library preparation. Targeted capture provides an efficient and sensitive means for sequencing specific genomic regions in a high-throughput manner. The libraries were sequenced to mean >85-100x coverage on Illumina Novaseq X sequencing platform with Paired End 2x150 chemistry.
- We follow the Illumina DRAGEN Bio-IT Platform for identification of variants in the sample. The sequences obtained are assembled and aligned to reference sequences based on NCBI RefSeq transcripts and human genome build (GRCh38).
- This assay (Whole exome sequencing) covers 20321 genes including 37 mitochondria genes.
- Haplotype caller has been used to identify variants which are relevant to the clinical indication.
- In addition to SNVs and small Indels, copy number variants (CNVs) are detected from targeted sequence data using the commercially available algorithm. This algorithm detects rare CNVs based on comparison of the read-depths of the test data with the matched aggregate reference dataset.
- Clinically relevant mutations were annotated using published variants in literature, Commercial datasets and a set of diseases databases. Common variants are filtered based on allele frequency in 1000Genome, ExAC, gnomAD, dbSNP & Laboratory's internal database.
- Non-synonymous variants effect is calculated using multiple algorithms such as PolyPhen-2, SIFT, MutationTaster2.
- Based on annotations data, ACMG rules-based classification performed for classification of variants identified through next generation sequencing study.

Limitations/Disclaimer:

- It should be noted that this test does not sequence all bases in a human genome, not all variants have been identified or interpreted, and this report is limited only to variants with evidence for causing or contributing to disease/clinical details provided to Laboratory.
- Testing has been performed assuming that the sample received belongs to the above named individual and any stated relationships between individuals are accepted as true.
- Test results are interpreted in the context of clinical findings, family history and other laboratory data. Only variations

information available. Re-interpretation of multi gene next generation sequencing data is recommended on an annual basis and may be requested by a medical provider.

- The sensitivity of this assay to detect large deletions/duplications of more than 10 bp or copy number variations (CNV) is 70-75%. The CNVs detected with this assay have to be confirmed by alternate method such as MLPA & Microarray.
- Due to inherent technology limitations, coverage is not uniform across all regions. Hence pathogenic variants present in areas of insufficient coverage as well as those variants which currently do not co-relate with the provided phenotype may not be analyzed/ reported. Additionally, it may not be possible to fully resolve certain details about variants, such as mosaicism, phasing, or mapping ambiguity.
- Triplet repeat expansions, translocations, large deletion or duplications and chromosomal rearrangements events are currently not reliably detected by next generation sequencing.
- This assay is not meant to interrogate most promoter regions, deep intronic regions, or other regulatory elements, other gene rearrangements like inversion or translocation and does not detect single or multi-exon deletions or duplications.
- Incidental or secondary findings (if any) that meet the ACMG guidelines can be given upon request.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- Sequence and copy number variants are reported according to the Human Genome Variation Society (HGVS).
- The transcript used for clinical reporting generally represents the canonical transcript, which is usually the longest coding transcript with strong/multiple supporting evidence. However, clinically relevant variants annotated in alternate complete coding transcripts could also be reported.

*Variant classification as per ACMG guidelines:

Variant	A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).
Benign	A variant which is known not to be responsible for disease has been detected. Generally, no further action is warranted on such variants when detected.
Likely Benign	A variant which is very unlikely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of Pathogenicity.
Pathogenic	A disease-causing variation in a gene which can explain the patient's symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed.

Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

***References*:**

1. Richards S, Aziz N, Bale S, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology, Genetics in Medicine, 2015 May;17(5):405-24

2. Kalia S.S. et al., Recommendations for reporting of secondary findings in clinical exome and genome sequencing, 2016 update(ACMG SF v2.0): a policy statement of the American College of Medical Genetics and Genomics. Genet Med., 19(2):249-255, 2017.

3. Meyer, L.R., et al., The UCSC Genome Browser database: extensions and updates 2013. Nucleic Acids Res, 2013. 41(D1): p.D64-9.

4. McKenna, A., et al., The Genome Analysis Toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. Genome Res, 2010. 20(9): p. 1297-303

5. 1000 Genomes Project Consortium et al., A global reference for human genetic variation. Nature, 526(7571): 68-74, 2015.

6. Lek M. et al., Analysis of protein-coding genetic variation in 60,706 humans. Nature, 536(7616):285-91, 2016.

7. McLaren, W., et al., Deriving the consequences of genomic variants with the Ensembl API and SNP Effect Predictor. Bioinformatics, 2010. 26(16): p. 2069-70.

Appendix 1: SAMPLE DATA AND STATISTICS

Data	
Total data generated (Gb)	7.54
Total Reads (%)	100.00
Total reads aligned (%)	98.83
Data >Q30 (%)	97.48

Appendix 3: MTDNA GENES

Gene	Region covered	Gene	Region covered	Gene	Region covered
MT-ATP8	100.00%	*MT-RNR1*	100.00%	*MT-TL2*	100.00%
MT-CO1	100.00%	*MT-RNR2*	100.00%	*MT-TM*	100.00%
MT-CO2	100.00%	*MT-TA*	100.00%	*MT-TN*	100.00%
MT-CO3	100.00%	*MT-TC*	100.00%	*MT-TP*	100.00%
MT-CYB	100.00%	*MT-TD*	100.00%	*MT-TQ*	100.00%
MT-ND1	100.00%	*MT-TE*	100.00%	*MT-TR*	100.00%
MT-ND2	100.00%	*MT-TF*	100.00%	*MT-TS1*	100.00%
MT-ND3	100.00%	*MT-TG*	100.00%	*MT-TS2*	100.00%
MT-ND4	100.00%	*MT-TH*	100.00%	*MT-TT*	100.00%
MT-ND4L	100.00%	*MT-TI*	100.00%	*MT-TV*	100.00%
MT-ND5	100.00%	*MT-TK*	100.00%	*MT-TW*	100.00%
MT-TY	100.00%				

Appendix 4: COVERAGE OF ANALYZED GENES (Percentage of coding region covered)

Gene	Region covered	Gene	Region covered	Gene	Region covered
AASS	100.00%	*ABAT*	100.00%	*ABCA1*	100.00%
ABCA7	100.00%	*ABCB6*	100.00%	*ABCB7*	100.00%
ABCC6	100.00%	*ABCC8*	100.00%	*ABCD1*	100.00%
ABCD3	100.00%	*ABCD4*	100.00%	*ABHD12*	100.00%
ABHD5	100.00%	*ACACA*	100.00%	*ACAD8*	100.00%
ACAD9	100.00%	*ACADM*	100.00%	*ACADS*	100.00%
ACADSB	100.00%	*ACADVL*	100.00%	*ACAT1*	100.00%
ACE	100.00%	*ACHE*	100.00%	*ACO2*	100.00%
ACOX1	100.00%	*ACSF3*	100.00%	*ACSL4*	100.00%
ACTA1	100.00%	*ACTA2*	100.00%	*ACTB*	100.00%
ACTG1	100.00%	*ACTG2*	100.00%	*ACTL6B*	100.00%
ACTN4	100.00%	*ACVRL1*	100.00%	*ACY1*	100.00%
ADA	100.00%	*ADAM10*	100.00%	*ADAM22*	100.00%

ADAMTS10	100.00%	ADAMTSL2	100.00%	ADAR	100.00%
AFF2	100.00%	*AFF3*	100.00%	*AFG3L2*	100.00%
AGA	100.00%	*AGK*	100.00%	*AGL*	100.00%
AGPS	100.00%	*AGRN*	100.00%	*AGTPBP1*	100.00%
AGXT	100.00%	*AHCY*	100.00%	*AHDC1*	100.00%
AHI1	100.00%	*AIFM1*	100.00%	*AIMP1*	100.00%
AIMP2	100.00%	*AK2*	100.00%	*AKT3*	100.00%
ALAD	100.00%	*ALAS2*	100.00%	*ALDH18A1*	100.00%
ALDH2	100.00%	*ALDH3A2*	100.00%	*ALDH4A1*	100.00%
ALDH5A1	100.00%	*ALDH6A1*	100.00%	*ALDH7A1*	100.00%
ALDOA	100.00%	*ALDOB*	100.00%	*ALG1*	100.00%
ALG11	100.00%	*ALG12*	100.00%	*ALG13*	100.00%
ALG14	100.00%	*ALG2*	100.00%	*ALG3*	100.00%
ALG6	100.00%	*ALG8*	100.00%	*ALG9*	100.00%
ALPL	100.00%	*ALS2*	100.00%	*ALX1*	100.00%
ALX3	100.00%	*ALX4*	100.00%	*AMACR*	100.00%
AMMECR1	100.00%	*AMPD1*	100.00%	*AMPD2*	100.00%
AMT	100.00%	*ANG*	100.00%	*ANK2*	100.00%
ANK3	100.00%	*ANKLE2*	100.00%	*ANKRD11*	100.00%
ANO10	100.00%	*ANO3*	100.00%	*ANO5*	100.00%
ANTXR2	100.00%	*ANXA11*	100.00%	*AP1S1*	100.00%
AP1S2	100.00%	*AP2M1*	100.00%	*AP3B1*	100.00%
AP3B2	100.00%	*AP4B1*	100.00%	*AP4E1*	100.00%
AP4M1	100.00%	*AP4S1*	100.00%	*AP5Z1*	100.00%
APOE	100.00%	*APP*	100.00%	*APTX*	100.00%
ARFGEF2	100.00%	*ARG1*	100.00%	*ARHGAP31*	100.00%
ARHGEF10	100.00%	*ARHGEF6*	100.00%	*ARHGEF9*	100.00%
ARID1A	100.00%	*ARID1B*	100.00%	*ARID2*	100.00%
ARL13B	100.00%	*ARL6*	100.00%	*ARL6IP1*	100.00%
ARSA	100.00%	*ARSB*	100.00%	*ARSL*	100.00%
ARV1	100.00%	*ARX*	100.00%	*ASAH1*	100.00%
ASCC1	100.00%	*ASCL1*	100.00%	*ASH1L*	100.00%
ASL	100.00%	*ASNS*	100.00%	*ASPA*	100.00%
ASPM	100.00%	*ASS1*	100.00%	*ASTN2*	100.00%

ASXL1	100.00%	ASXL3	100.00%	ATAD1	100.00%
ATP2A1	100.00%	*ATP2A2*	100.00%	*ATP2B3*	100.00%
ATP2B4	100.00%	*ATP5F1A*	100.00%	*ATP5F1E*	100.00%
ATP6AP1	100.00%	*ATP6AP2*	100.00%	*ATP6V0A2*	100.00%
ATP6V1A	100.00%	*ATP7A*	100.00%	*ATP7B*	100.00%
ATP8A2	100.00%	*ATPAF2*	100.00%	*ATR*	100.00%
ATRX	100.00%	*AUH*	100.00%	*AUTS2*	100.00%
B3GALNT2	100.00%	*B3GLCT*	100.00%	*B4GALNT1*	100.00%
B4GALT1	100.00%	*B4GAT1*	100.00%	*B9D1*	100.00%
B9D2	100.00%	*BAG3*	100.00%	*BBS1*	100.00%
BBS10	100.00%	*BBS12*	100.00%	*BBS2*	100.00%
BBS4	100.00%	*BBS5*	100.00%	*BBS7*	100.00%
BBS9	100.00%	*BCAP31*	100.00%	*BCKDHA*	100.00%
BCKDHB	100.00%	*BCKDK*	100.00%	*BCL11A*	100.00%
BCOR	100.00%	*BCS1L*	100.00%	*BDNF*	100.00%
BEST1	100.00%	*BICD2*	100.00%	*BIN1*	100.00%
BLOC1S1	100.00%	*BLOC1S3*	100.00%	*BLOC1S6*	100.00%
BOLA3	100.00%	*BRAF*	100.00%	*BRAT1*	100.00%
BRWD3	100.00%	*BSCL2*	100.00%	*BSND*	100.00%
BTD	100.00%	*BVES*	100.00%	*C12orf4*	100.00%
C12orf57	100.00%	*C19orf12*	100.00%	*C1QBP*	100.00%
CA2	100.00%	*CA5A*	100.00%	*CA8*	100.00%
CACNA1A	100.00%	*CACNA1B*	100.00%	*CACNA1C*	100.00%
CACNA1D	100.00%	*CACNA1E*	100.00%	*CACNA1F*	100.00%
CACNA1G	100.00%	*CACNA1H*	100.00%	*CACNA1S*	100.00%
CACNA2D2	100.00%	*CACNB2*	100.00%	*CACNB4*	100.00%
CAD	100.00%	*CAMK2A*	100.00%	*CAMK2B*	100.00%
CAMK2G	100.00%	*CAMTA1*	100.00%	*CAPN1*	100.00%
CAPN3	100.00%	*CARD11*	100.00%	*CARS2*	100.00%
CASK	100.00%	*CASQ1*	100.00%	*CASR*	100.00%
CAT	100.00%	*CAV1*	100.00%	*CAV3*	100.00%
CAVIN1	100.00%	*CBL*	100.00%	*CBS*	100.00%
CC2D1A	100.00%	*CC2D2A*	100.00%	*CCDC115*	100.00%
CCDC22	100.00%	*CCDC40*	100.00%	*CCDC78*	100.00%

CCDC88A	100.00%	CCDC88C	100.00%	CCM2	100.00%
CDON	100.00%	*CEL*	100.00%	*CENPF*	100.00%
CENPJ	100.00%	*CEP135*	100.00%	*CEP152*	100.00%
CEP164	100.00%	*CEP290*	100.00%	*CEP41*	100.00%
CEP63	100.00%	*CERS1*	100.00%	*CERT1*	100.00%
CFAP418	100.00%	*CFL2*	100.00%	*CHAMP1*	100.00%
CHAT	100.00%	*CHCHD10*	100.00%	*CHCHD2*	100.00%
CHD1	100.00%	*CHD2*	100.00%	*CHD3*	100.00%
CHD7	100.00%	*CHD8*	100.00%	*CHKB*	100.00%
CHL1	100.00%	*CHMP1A*	100.00%	*CHMP2B*	100.00%
CHRNA1	100.00%	*CHRNA2*	100.00%	*CHRNA4*	100.00%
CHRNA7	100.00%	*CHRNB1*	100.00%	*CHRNB2*	100.00%
CHRND	100.00%	*CHRNE*	100.00%	*CHRNG*	100.00%
CHST14	100.00%	*CHSY1*	100.00%	*CIB2*	100.00%
CIC	100.00%	*CILK1*	100.00%	*CISD2*	100.00%
CIT	100.00%	*CLCN1*	100.00%	*CLCN2*	100.00%
CLCN4	100.00%	*CLCNKA*	100.00%	*CLCNKB*	100.00%
CLDN16	100.00%	*CLDN19*	100.00%	*CLN3*	100.00%
CLN5	100.00%	*CLN6*	100.00%	*CLN8*	100.00%
CLP1	100.00%	*CLPB*	100.00%	*CLPP*	100.00%
CLTC	100.00%	*CNBP*	100.00%	*CNGB3*	100.00%
CNKSR2	100.00%	*CNNM2*	100.00%	*CNPY3*	100.00%
CNTN4	100.00%	*CNTNAP1*	100.00%	*CNTNAP2*	100.00%
COA5	100.00%	*COA6*	100.00%	*COA7*	100.00%
COA8	100.00%	*COASY*	100.00%	*COG1*	100.00%
COG4	100.00%	*COG5*	100.00%	*COG6*	100.00%
COG7	100.00%	*COG8*	100.00%	*COL11A2*	100.00%
COL12A1	100.00%	*COL13A1*	100.00%	*COL2A1*	100.00%
COL4A1	100.00%	*COL4A2*	100.00%	*COL6A1*	100.00%
COL6A2	100.00%	*COL6A3*	100.00%	*COLGALT1*	100.00%
COLQ	100.00%	*COMT*	100.00%	*COQ2*	100.00%
COQ4	100.00%	*COQ6*	100.00%	*COQ7*	100.00%
COQ8A	100.00%	*COQ8B*	100.00%	*COQ9*	100.00%
COX10	100.00%	*COX14*	100.00%	*COX15*	100.00%

COX20	100.00%	COX4I2	100.00%	COX6A1	100.00%
CPT2	100.00%	*CRADD*	100.00%	*CRBN*	100.00%
CREBBP	100.00%	*CRIPT*	100.00%	*CRLF1*	100.00%
CRPPA	100.00%	*CRYAB*	100.00%	*CSF1R*	100.00%
CSMD1	100.00%	*CSNK2B*	100.00%	*CSPP1*	100.00%
CSRP3	100.00%	*CST3*	100.00%	*CSTB*	100.00%
CTC1	100.00%	*CTCF*	100.00%	*CTDP1*	100.00%
CTNNA2	100.00%	*CTNNA3*	100.00%	*CTNNB1*	100.00%
CTNS	100.00%	*CTSA*	100.00%	*CTSC*	100.00%
CTSD	100.00%	*CTSF*	100.00%	*CTSK*	100.00%
CUL3	100.00%	*CUL4B*	100.00%	*CUL7*	100.00%
CUX1	100.00%	*CUX2*	100.00%	*CWF19L1*	100.00%
CX3CR1	100.00%	*CYB5A*	100.00%	*CYB5R3*	100.00%
CYC1	100.00%	*CYCS*	100.00%	*CYFIP2*	100.00%
CYLD	100.00%	*CYP11A1*	100.00%	*CYP11B1*	100.00%
CYP11B2	100.00%	*CYP24A1*	100.00%	*CYP27A1*	100.00%
CYP27B1	100.00%	*CYP2U1*	100.00%	*CYP7B1*	100.00%
D2HGDH	100.00%	*DAB1*	100.00%	*DAG1*	100.00%
DARS1	100.00%	*DARS2*	100.00%	*DBT*	100.00%
DCAF17	100.00%	*DCTN1*	100.00%	*DCX*	100.00%
DDC	100.00%	*DDHD1*	100.00%	*DDHD2*	100.00%
DDOST	100.00%	*DDX3X*	100.00%	*DEAF1*	100.00%
DEGS1	100.00%	*DENND5A*	100.00%	*DEPDC5*	100.00%
DES	100.00%	*DGUOK*	100.00%	*DHCR24*	100.00%
DHCR7	100.00%	*DHDDS*	100.00%	*DHFR*	100.00%
DHH	100.00%	*DHODH*	100.00%	*DHPS*	100.00%
DHTKD1	100.00%	*DHX30*	100.00%	*DIABLO*	100.00%
DIAPH1	100.00%	*DIAPH3*	100.00%	*DIP2B*	100.00%
DKC1	100.00%	*DLAT*	100.00%	*DLD*	100.00%
DLG3	100.00%	*DLG4*	100.00%	*DLGAP2*	100.00%
DLL3	100.00%	*DLX3*	100.00%	*DMD*	100.00%
DMGDH	100.00%	*DMPK*	100.00%	*DMXL2*	100.00%
DNA2	100.00%	*DNAJB2*	100.00%	*DNAJB6*	100.00%
DNAJC12	100.00%	*DNAJC13*	100.00%	*DNAJC19*	100.00%

DNAJC5	100.00%	DNAJC6	100.00%	DNM1	100.00%
DOLK	100.00%	*DPAGT1*	100.00%	*DPF2*	100.00%
DPM1	100.00%	*DPM2*	100.00%	*DPM3*	100.00%
DPP6	100.00%	*DPYD*	100.00%	*DPYS*	100.00%
DRD3	100.00%	*DST*	100.00%	*DSTYK*	100.00%
DTNBP1	100.00%	*DVL3*	100.00%	*DYM*	100.00%
DYNC1H1	100.00%	*DYNC2H1*	100.00%	*DYRK1A*	100.00%
DYSF	100.00%	*EARS2*	100.00%	*EBF3*	100.00%
EBP	100.00%	*ECEL1*	100.00%	*ECHS1*	100.00%
EDC3	100.00%	*EDN3*	100.00%	*EDNRB*	100.00%
EEF1A2	100.00%	*EFHC1*	100.00%	*EFTUD2*	100.00%
EGF	100.00%	*EGR2*	100.00%	*EHMT1*	100.00%
EIF2B1	100.00%	*EIF2B2*	100.00%	*EIF2B3*	100.00%
EIF2B4	100.00%	*EIF2B5*	100.00%	*EIF2S3*	100.00%
EIF3F	100.00%	*EIF4G1*	100.00%	*ELAC2*	100.00%
ELOVL4	100.00%	*ELOVL5*	100.00%	*ELP1*	100.00%
ELP2	100.00%	*EMC10*	100.00%	*EMD*	100.00%
EML1	100.00%	*EMX2*	100.00%	*ENO3*	100.00%
ENTPD1	100.00%	*EP300*	100.00%	*EPB41L1*	100.00%
EPG5	100.00%	*EPHX2*	100.00%	*EPM2A*	100.00%
EPRS1	100.00%	*ERBB4*	100.00%	*ERCC1*	100.00%
ERCC2	100.00%	*ERCC5*	100.00%	*ERCC6*	100.00%
ERCC8	100.00%	*ERLIN1*	100.00%	*ERLIN2*	100.00%
ESCO2	100.00%	*ETFA*	100.00%	*ETFB*	100.00%
ETFDH	100.00%	*ETHE1*	100.00%	*EWSR1*	100.00%
EXOC6B	100.00%	*EXOSC3*	100.00%	*EXOSC8*	100.00%
EXOSC9	100.00%	*EXT1*	100.00%	*EZH2*	100.00%
F2	100.00%	*F5*	100.00%	*FA2H*	100.00%
FADD	100.00%	*FAH*	100.00%	*FAM126A*	100.00%
FAN1	100.00%	*FANCB*	100.00%	*FARS2*	100.00%
FARSB	100.00%	*FASTKD2*	100.00%	*FAT2*	100.00%
FBLN5	100.00%	*FBN1*	100.00%	*FBN2*	100.00%
FBXL4	100.00%	*FBXO11*	100.00%	*FBXO38*	100.00%
FBXO7	100.00%	*FDX2*	100.00%	*FDXR*	100.00%

FECH	100.00%	FEZF1	100.00%	FGA	100.00%
FIG4	100.00%	*FKBP10*	100.00%	*FKBP14*	100.00%
FKRP	100.00%	*FKTN*	100.00%	*FLAD1*	100.00%
FLNA	100.00%	*FLNC*	100.00%	*FLVCR1*	100.00%
FLVCR2	100.00%	*FMN2*	100.00%	*FOLR1*	100.00%
FOXC1	100.00%	*FOXG1*	100.00%	*FOXL2*	100.00%
FOXP1	100.00%	*FOXP2*	100.00%	*FOXRED1*	100.00%
FRMD7	100.00%	*FRMPD4*	100.00%	*FRRS1L*	100.00%
FTL	100.00%	*FTO*	100.00%	*FTSJ1*	100.00%
FUCA1	100.00%	*FUS*	100.00%	*FUT8*	100.00%
FXN	100.00%	*FXR1*	100.00%	*FXYD2*	100.00%
G6PD	100.00%	*GAA*	100.00%	*GABBR2*	100.00%
GABRA1	100.00%	*GABRA2*	100.00%	*GABRA5*	100.00%
GABRB1	100.00%	*GABRB2*	100.00%	*GABRB3*	100.00%
GABRD	100.00%	*GABRE*	100.00%	*GABRG2*	100.00%
GAD1	100.00%	*GALC*	100.00%	*GALNS*	100.00%
GALT	100.00%	*GAMT*	100.00%	*GAN*	100.00%
GARS1	100.00%	*GATAD2B*	100.00%	*GATM*	100.00%
GBA	100.00%	*GBA2*	100.00%	*GBE1*	100.00%
GCDH	100.00%	*GCH1*	100.00%	*GCK*	100.00%
GCSH	100.00%	*GDAP1*	100.00%	*GDI1*	100.00%
GDNF	100.00%	*GFAP*	100.00%	*GFER*	100.00%
GFM1	100.00%	*GFM2*	100.00%	*GFPT1*	100.00%
GIGYF2	100.00%	*GJA1*	100.00%	*GJB1*	100.00%
GJB3	100.00%	*GJC2*	100.00%	*GK*	100.00%
GLA	100.00%	*GLB1*	100.00%	*GLDC*	100.00%
GLDN	100.00%	*GLE1*	100.00%	*GLI2*	100.00%
GLI3	100.00%	*GLO1*	100.00%	*GLRA1*	100.00%
GLRB	100.00%	*GLRX5*	100.00%	*GLUD1*	100.00%
GLUL	100.00%	*GLYCTK*	100.00%	*GM2A*	100.00%
GMPPA	100.00%	*GMPPB*	100.00%	*GNAL*	100.00%
GNAO1	100.00%	*GNAQ*	100.00%	*GNAS*	100.00%
GNB1	100.00%	*GNB4*	100.00%	*GNB5*	100.00%
GNE	100.00%	*GNPAT*	100.00%	*GNPTAB*	100.00%

GNPTG	100.00%	GNS	100.00%	GOSR2	100.00%
GRHPR	100.00%	*GRIA1*	100.00%	*GRIA2*	100.00%
GRIA3	100.00%	*GRIA4*	100.00%	*GRID2*	100.00%
GRIK2	100.00%	*GRIN1*	100.00%	*GRIN2A*	100.00%
GRIN2B	100.00%	*GRIN2D*	100.00%	*GRIP1*	100.00%
GRM1	100.00%	*GRN*	100.00%	*GSN*	100.00%
GSR	100.00%	*GSS*	100.00%	*GTPBP2*	100.00%
GTPBP3	100.00%	*GUF1*	100.00%	*GUSB*	100.00%
GYG1	100.00%	*GYS1*	100.00%	*HACE1*	100.00%
HADH	100.00%	*HADHA*	100.00%	*HADHB*	100.00%
HAMP	100.00%	*HARS2*	100.00%	*HAX1*	100.00%
HBB	100.00%	*HCCS*	100.00%	*HCFC1*	100.00%
HCN1	100.00%	*HDAC4*	100.00%	*HDAC8*	100.00%
HECW2	100.00%	*HEPACAM*	100.00%	*HERC2*	100.00%
HESX1	100.00%	*HEXA*	100.00%	*HEXB*	100.00%
HGSNAT	100.00%	*HIBCH*	100.00%	*HIKESHI*	100.00%
HINT1	100.00%	*HIVEP2*	100.00%	*HK1*	100.00%
HLCS	100.00%	*HMBS*	100.00%	*HMGCL*	100.00%
HMGCS2	100.00%	*HNMT*	100.00%	*HNRNPA1*	100.00%
HNRNPA2B1	100.00%	*HNRNPDL*	100.00%	*HNRNPH2*	100.00%
HNRNPR	100.00%	*HNRNPU*	100.00%	*HOGA1*	100.00%
HOXA1	100.00%	*HOXD10*	100.00%	*HPCA*	100.00%
HPD	100.00%	*HPRT1*	100.00%	*HPS1*	100.00%
HPS4	100.00%	*HPS5*	100.00%	*HPS6*	100.00%
HRAS	100.00%	*HSD11B1*	100.00%	*HSD17B10*	100.00%
HSD17B4	100.00%	*HSD3B2*	100.00%	*HSPA9*	100.00%
HSPB1	100.00%	*HSPB3*	100.00%	*HSPB8*	100.00%
HSPD1	100.00%	*HSPG2*	100.00%	*HTRA1*	100.00%
HTRA2	100.00%	*HUWE1*	100.00%	*HYAL1*	100.00%
HYDIN	100.00%	*IARS2*	100.00%	*IBA57*	100.00%
IDH2	100.00%	*IDH3B*	100.00%	*IDS*	100.00%
IDUA	100.00%	*IER3IP1*	100.00%	*IFIH1*	100.00%
IFT140	100.00%	*IFT172*	100.00%	*IFT27*	100.00%
IGBP1	100.00%	*IGF1*	100.00%	*IGF1R*	100.00%

IGHMBP2	100.00%	IL1RAPL1	100.00%	IMPA1	100.00%
ITGA7	100.00%	*ITGB3*	100.00%	*ITM2B*	100.00%
ITPA	100.00%	*ITPR1*	100.00%	*IVD*	100.00%
JAG1	100.00%	*JAM2*	100.00%	*JAM3*	100.00%
KANK1	100.00%	*KANSL1*	100.00%	*KARS1*	100.00%
KAT6A	100.00%	*KAT6B*	100.00%	*KAT8*	100.00%
KATNB1	100.00%	*KBTBD13*	100.00%	*KCNA1*	100.00%
KCNA2	100.00%	*KCNB1*	100.00%	*KCNC1*	100.00%
KCNC3	100.00%	*KCND3*	100.00%	*KCNE3*	100.00%
KCNH1	100.00%	*KCNJ1*	100.00%	*KCNJ10*	100.00%
KCNJ2	100.00%	*KCNK18*	100.00%	*KCNK4*	100.00%
KCNK9	100.00%	*KCNMA1*	100.00%	*KCNQ2*	100.00%
KCNQ3	100.00%	*KCNQ5*	100.00%	*KCNT1*	100.00%
KCNT2	100.00%	*KCTD17*	100.00%	*KCTD3*	100.00%
KCTD7	100.00%	*KDM4B*	100.00%	*KDM5B*	100.00%
KDM5C	100.00%	*KDM6A*	100.00%	*BLTP1*	100.00%
KIDINS220	100.00%	*KIF11*	100.00%	*KIF14*	100.00%
KIF1A	100.00%	*KIF1B*	100.00%	*KIF1C*	100.00%
KIF21A	100.00%	*KIF2A*	100.00%	*KIF5A*	100.00%
KIF5C	100.00%	*KIF7*	100.00%	*KIFBP*	100.00%
KIRREL3	100.00%	*KLHL40*	100.00%	*KLHL41*	100.00%
KLHL7	100.00%	*KMT2A*	100.00%	*KMT2B*	100.00%
KMT2C	100.00%	*KMT2D*	100.00%	*KMT2E*	100.00%
KMT5B	100.00%	*KNL1*	100.00%	*KPTN*	100.00%
KRAS	100.00%	*KRIT1*	100.00%	*KRT5*	100.00%
KRT8	100.00%	*KY*	100.00%	*L1CAM*	100.00%
L2HGDH	100.00%	*LAMA1*	100.00%	*LAMA2*	100.00%
LAMB1	100.00%	*LAMB2*	100.00%	*LAMC3*	100.00%
LAMP2	100.00%	*LARGE1*	100.00%	*LARS2*	100.00%
LAT	100.00%	*LBR*	100.00%	*LDB3*	100.00%
LDHA	100.00%	*LEP*	100.00%	*LGI1*	100.00%
LGI4	100.00%	*LHX3*	100.00%	*LHX4*	100.00%
LIAS	100.00%	*LIMS2*	100.00%	*LINS1*	100.00%
LIPA	100.00%	*LIPT1*	100.00%	*LIPT2*	100.00%

LITAF	100.00%	LMAN2L	100.00%	LMBRD1	100.00%
LRP2	100.00%	*LRP4*	100.00%	*LRPPRC*	100.00%
LRRK2	100.00%	*LRSAM1*	100.00%	*LYRM7*	100.00%
LYST	100.00%	*LZTFL1*	100.00%	*LZTR1*	100.00%
MACF1	100.00%	*MAF*	100.00%	*MAG*	100.00%
MAGEL2	100.00%	*MAGI2*	100.00%	*MAGT1*	100.00%
MAMLD1	100.00%	*MAN1B1*	100.00%	*MAN2B1*	100.00%
MANBA	100.00%	*MAOA*	100.00%	*MAP2K1*	100.00%
MAP2K2	100.00%	*MAPT*	100.00%	*MARS1*	100.00%
MARS2	100.00%	*MASP1*	100.00%	*MATR3*	100.00%
MBD5	100.00%	*MBOAT7*	100.00%	*MBTPS2*	100.00%
MCCC1	100.00%	*MCCC2*	100.00%	*MCEE*	100.00%
MCM4	100.00%	*MCM6*	100.00%	*MCOLN1*	100.00%
MCPH1	100.00%	*MDH2*	100.00%	*MECP2*	100.00%
MECR	100.00%	*MED12*	100.00%	*MED13*	100.00%
MED13L	100.00%	*MED17*	100.00%	*MED23*	100.00%
MED25	100.00%	*MEF2C*	100.00%	*MEGF10*	100.00%
MEIS2	100.00%	*METTL23*	100.00%	*MFF*	100.00%
MFN2	100.00%	*MFRP*	100.00%	*MFSD2A*	100.00%
MFSD8	100.00%	*MGAT2*	100.00%	*MGME1*	100.00%
MIB1	100.00%	*MICU1*	100.00%	*MID1*	100.00%
MIPEP	100.00%	*MITF*	100.00%	*MKKS*	100.00%
MKS1	100.00%	*MLC1*	100.00%	*MLPH*	100.00%
MLYCD	100.00%	*MMAA*	100.00%	*MMAB*	100.00%
MMACHC	100.00%	*MMADHC*	100.00%	*MMUT*	100.00%
MOCS1	100.00%	*MOCS2*	100.00%	*MOGS*	100.00%
MPC1	100.00%	*MPDU1*	100.00%	*MPDZ*	100.00%
MPI	100.00%	*MPV17*	100.00%	*MPZ*	100.00%
MRE11	100.00%	*MRPL3*	100.00%	*MRPL44*	100.00%
MRPS16	100.00%	*MRPS2*	100.00%	*MRPS22*	100.00%
MRPS34	100.00%	*MSMO1*	100.00%	*MSR1*	100.00%
MSRB3	100.00%	*MSTO1*	100.00%	*MSX1*	100.00%
MSX2	100.00%	*MTFMT*	100.00%	*MTHFD1*	100.00%
MTHFR	100.00%	*MTHFS*	100.00%	*MTM1*	100.00%

MTMR14	100.00%	MTMR2	100.00%	MTO1	100.00%
MYCN	100.00%	*MYH2*	100.00%	*MYH3*	100.00%
MYH7	100.00%	*MYH8*	100.00%	*MYL1*	100.00%
MYL2	100.00%	*MYMK*	100.00%	*MYO18B*	100.00%
MYO1E	100.00%	*MYO5A*	100.00%	*MYO9A*	100.00%
MYO9B	100.00%	*MYORG*	100.00%	*MYOT*	100.00%
MYPN	100.00%	*MYT1L*	100.00%	*NAA10*	100.00%
NAA15	100.00%	*NACC1*	100.00%	*NADK2*	100.00%
NAGA	100.00%	*NAGLU*	100.00%	*NAGS*	100.00%
NALCN	100.00%	*NARS2*	100.00%	*NAXD*	100.00%
NAXE	100.00%	*NBAS*	100.00%	*NBEA*	100.00%
NCAPD3	100.00%	*NDE1*	100.00%	*NDP*	100.00%
NDRG1	100.00%	*NDST1*	100.00%	*NDUFA1*	100.00%
NDUFA10	100.00%	*NDUFA11*	100.00%	*NDUFA12*	100.00%
NDUFA2	100.00%	*NDUFA6*	100.00%	*NDUFA9*	100.00%
NDUFAF1	100.00%	*NDUFAF2*	100.00%	*NDUFAF3*	100.00%
NDUFAF4	100.00%	*NDUFAF5*	100.00%	*NDUFAF6*	100.00%
NDUFB11	100.00%	*NDUFB3*	100.00%	*NDUFB8*	100.00%
NDUFB9	100.00%	*NDUFS1*	100.00%	*NDUFS2*	100.00%
NDUFS3	100.00%	*NDUFS4*	100.00%	*NDUFS6*	100.00%
NDUFS7	100.00%	*NDUFS8*	100.00%	*NDUFV1*	100.00%
NDUFV2	100.00%	*NEB*	100.00%	*NECAP1*	100.00%
NECTIN1	100.00%	*NEDD4L*	100.00%	*NEFH*	100.00%
NEK1	100.00%	*NEK10*	100.00%	*NEU1*	100.00%
NEUROD2	100.00%	*NEXMIF*	100.00%	*NF1*	100.00%
NFE2L2	100.00%	*NFIA*	100.00%	*NFIB*	100.00%
NFIX	100.00%	*NFU1*	100.00%	*NGF*	100.00%
NGLY1	100.00%	*NHEJ1*	100.00%	*NHLRC1*	100.00%
NHS	100.00%	*NIPA1*	100.00%	*NIPBL*	100.00%
NKX6-2	100.00%	*NLGN3*	100.00%	*NLGN4X*	100.00%
NLRP12	100.00%	*NLRP3*	100.00%	*NNT*	100.00%
NOTCH1	100.00%	*NOTCH2*	100.00%	*NOTCH3*	100.00%
NPC1	100.00%	*NPC2*	100.00%	*NPHP1*	100.00%
NPHP3	100.00%	*NPR2*	100.00%	*NPRL2*	100.00%

NPRL3	100.00%	NR2F1	100.00%	NR3C2	100.00%
NTRK2	100.00%	*NUBPL*	100.00%	*NUP133*	100.00%
NUP62	100.00%	*NUS1*	100.00%	*OAT*	100.00%
OCLN	100.00%	*OCRL*	100.00%	*ODAD4*	100.00%
OFD1	100.00%	*OGDH*	100.00%	*OPA1*	100.00%
OPA3	100.00%	*OPHN1*	100.00%	*OPTN*	100.00%
ORC1	100.00%	*OSGEP*	100.00%	*OTC*	100.00%
OTUD6B	100.00%	*OXCT1*	100.00%	*P4HA2*	100.00%
P4HB	100.00%	*P4HTM*	100.00%	*PACS1*	100.00%
PACS2	100.00%	*PAFAH1B1*	100.00%	*PAH*	100.00%
PAK1	100.00%	*PAK3*	100.00%	*PAM16*	100.00%
PANK2	100.00%	*PARK7*	100.00%	*PARS2*	100.00%
PAX3	100.00%	*PAX6*	100.00%	*PAX7*	100.00%
PBX1	100.00%	*PC*	100.00%	*PCBD1*	100.00%
PCCA	100.00%	*PCCB*	100.00%	*PCDH12*	100.00%
PCDH15	100.00%	*PCDH19*	100.00%	*PCK2*	100.00%
PCNT	100.00%	*PCYT2*	100.00%	*PDCD1*	100.00%
PDCD10	100.00%	*PDE10A*	100.00%	*PDE6D*	100.00%
PDE8B	100.00%	*PDGFB*	100.00%	*PDGFRB*	100.00%
PDHA1	100.00%	*PDHB*	100.00%	*PDHX*	100.00%
PDK3	100.00%	*PDP1*	100.00%	*PDSS1*	100.00%
PDSS2	100.00%	*PDX1*	100.00%	*PDYN*	100.00%
PER2	100.00%	*PET100*	100.00%	*PEX1*	100.00%
PEX10	100.00%	*PEX11B*	100.00%	*PEX12*	100.00%
PEX13	100.00%	*PEX14*	100.00%	*PEX16*	100.00%
PEX19	100.00%	*PEX2*	100.00%	*PEX26*	100.00%
PEX3	100.00%	*PEX5*	100.00%	*PEX6*	100.00%
PEX7	100.00%	*PFKM*	100.00%	*PFN1*	100.00%
PGAM2	100.00%	*PGAP1*	100.00%	*PGAP2*	100.00%
PGK1	100.00%	*PGM1*	100.00%	*PHACTR1*	100.00%
PHF6	100.00%	*PHF8*	100.00%	*PHGDH*	100.00%
PHIP	100.00%	*PHKA1*	100.00%	*PHOX2B*	100.00%
PHYH	100.00%	*PIEZO2*	100.00%	*PIGA*	100.00%
PIGB	100.00%	*PIGC*	100.00%	*PIGG*	100.00%

PIGH	100.00%	PIGL	100.00%	PIGN	100.00%
PIK3R2	100.00%	*PIK3R5*	100.00%	*PINK1*	100.00%
PIP5K1C	100.00%	*PITX1*	100.00%	*PITX2*	100.00%
PKLR	100.00%	*PLA2G6*	100.00%	*PLAA*	100.00%
PLCB1	100.00%	*PLCG2*	100.00%	*PLEC*	100.00%
PLEKHG2	100.00%	*PLEKHG5*	100.00%	*PLK1*	100.00%
PLK4	100.00%	*PLN*	100.00%	*PLOD2*	100.00%
PLP1	100.00%	*PLPBP*	100.00%	*PLXNB3*	100.00%
PMM2	100.00%	*PMP22*	100.00%	*PMPCA*	100.00%
PMPCB	100.00%	*PNKD*	100.00%	*PNKP*	100.00%
PNPLA2	100.00%	*PNPLA6*	100.00%	*PNPLA8*	100.00%
PNPO	100.00%	*PNPT1*	100.00%	*POGLUT1*	100.00%
POGZ	100.00%	*POLA1*	100.00%	*POLG*	100.00%
POLG2	100.00%	*POLR1C*	100.00%	*POLR1D*	100.00%
POLR3A	100.00%	*POLR3B*	100.00%	*POMGNT1*	100.00%
POMGNT2	100.00%	*POMK*	100.00%	*POMT1*	100.00%
POMT2	100.00%	*PON1*	100.00%	*POP1*	100.00%
PORCN	100.00%	*POT1*	100.00%	*POU1F1*	100.00%
PPM1D	100.00%	*PPOX*	100.00%	*PPP2CA*	100.00%
PPP2R1A	100.00%	*PPP2R5D*	100.00%	*PPP3CA*	100.00%
PPT1	100.00%	*PQBP1*	100.00%	*PREPL*	100.00%
PRF1	100.00%	*PRICKLE1*	100.00%	*PRICKLE2*	100.00%
PRKAG2	100.00%	*PRKCA*	100.00%	*PRKCG*	100.00%
PRKN	100.00%	*PRKRA*	100.00%	*PRMT7*	100.00%
PRNP	100.00%	*PRODH*	100.00%	*PROP1*	100.00%
PRPH	100.00%	*PRPS1*	100.00%	*PRRT2*	100.00%
PRRX1	100.00%	*PRSS12*	100.00%	*PRX*	100.00%
PSAP	100.00%	*PSAT1*	100.00%	*PSEN1*	100.00%
PSEN2	100.00%	*PSMD12*	100.00%	*PSPH*	100.00%
PTCH1	100.00%	*PTCHD1*	100.00%	*PTEN*	100.00%
PTF1A	100.00%	*PTPN11*	100.00%	*PTPN23*	100.00%
PTPRC	100.00%	*PTRH2*	100.00%	*PTS*	100.00%
PUM1	100.00%	*PURA*	100.00%	*PUS1*	100.00%
PUS3	100.00%	*PXDN*	100.00%	*PYCR1*	100.00%

PYCR2	100.00%	PYGM	100.00%	PYROXD1	100.00%
RAB3GAP2	100.00%	*RAB7A*	100.00%	*RAC1*	100.00%
RAD21	100.00%	*RAD50*	100.00%	*RAF1*	100.00%
RAI1	100.00%	*RALA*	100.00%	*RALGAPA1*	100.00%
RAP1GDS1	100.00%	*RAPSN*	100.00%	*RARS1*	100.00%
RARS2	100.00%	*RBBP8*	100.00%	*RBCK1*	100.00%
RBFOX1	100.00%	*RBM10*	100.00%	*RBM8A*	100.00%
RDH11	100.00%	*REEP1*	100.00%	*REEP2*	100.00%
RELN	100.00%	*RERE*	100.00%	*REST*	100.00%
RET	100.00%	*RETREG1*	100.00%	*RFT1*	100.00%
RHOBTB2	100.00%	*RIMS1*	100.00%	*RIN2*	100.00%
RMND1	100.00%	*RNASEH1*	100.00%	*RNASEH2A*	100.00%
RNASEH2B	100.00%	*RNASEH2C*	100.00%	*RNASET2*	100.00%
RNF113A	100.00%	*RNF13*	100.00%	*RNF135*	100.00%
RNF168	100.00%	*RNF170*	100.00%	*RNF216*	100.00%
ROBO2	100.00%	*ROGDI*	100.00%	*ROR2*	100.00%
RORA	100.00%	*RORB*	100.00%	*RPGRIP1L*	100.00%
RPIA	100.00%	*RPL10*	100.00%	*RPL35A*	100.00%
RPS14	100.00%	*RPS6KA3*	100.00%	*RRM2B*	100.00%
RTN2	100.00%	*RTN4IP1*	100.00%	*RTTN*	100.00%
RUBCN	100.00%	*RUSC2*	100.00%	*RXYLT1*	100.00%
RYR1	100.00%	*SACS*	100.00%	*SALL1*	100.00%
SAMD9L	100.00%	*SAMHD1*	100.00%	*SARS2*	100.00%
SASH1	100.00%	*SASS6*	100.00%	*SATB2*	100.00%
SBDS	100.00%	*SBF1*	100.00%	*SBF2*	100.00%
SC5D	100.00%	*SCARB2*	100.00%	*SCN10A*	100.00%
SCN1A	100.00%	*SCN1B*	100.00%	*SCN2A*	100.00%
SCN3A	100.00%	*SCN4A*	100.00%	*SCN8A*	100.00%
SCN9A	100.00%	*SCO1*	100.00%	*SCO2*	100.00%
SCYL1	100.00%	*SDCCAG8*	100.00%	*SDHA*	100.00%
SDHAF1	100.00%	*SDHAF2*	100.00%	*SDHB*	100.00%
SDHD	100.00%	*SEC23B*	100.00%	*SECISBP2*	100.00%
SELENOI	100.00%	*SELENON*	100.00%	*SEMA5A*	100.00%
SEMA6B	100.00%	*SEPSECS*	100.00%	*SERAC1*	100.00%

SERPINI1	**100.00%**	**SET**	**100.00%**	**SETBP1**	**100.00%**
SGCE	100.00%	*SGCG*	100.00%	*SGSH*	100.00%
SH3TC2	100.00%	*SHANK2*	100.00%	*SHH*	100.00%
SHOC2	100.00%	*SHROOM4*	100.00%	*SIGMAR1*	100.00%
SIK1	100.00%	*SIL1*	100.00%	*SIN3A*	100.00%
SIX3	100.00%	*SKI*	100.00%	*SLC12A3*	100.00%
SLC12A5	100.00%	*SLC12A6*	100.00%	*SLC13A3*	100.00%
SLC13A5	100.00%	*SLC16A1*	100.00%	*SLC16A2*	100.00%
SLC17A5	100.00%	*SLC18A3*	100.00%	*SLC19A2*	100.00%
SLC19A3	100.00%	*SLC1A1*	100.00%	*SLC1A2*	100.00%
SLC1A3	100.00%	*SLC1A4*	100.00%	*SLC20A2*	100.00%
SLC22A5	100.00%	*SLC25A1*	100.00%	*SLC25A12*	100.00%
SLC25A13	100.00%	*SLC25A15*	100.00%	*SLC25A19*	100.00%
SLC25A20	100.00%	*SLC25A22*	100.00%	*SLC25A26*	100.00%
SLC25A3	100.00%	*SLC25A38*	100.00%	*SLC25A4*	100.00%
SLC25A42	100.00%	*SLC25A46*	100.00%	*SLC27A4*	100.00%
SLC2A1	100.00%	*SLC2A10*	100.00%	*SLC30A10*	100.00%
SLC33A1	100.00%	*SLC35A1*	100.00%	*SLC35A2*	100.00%
SLC35A3	100.00%	*SLC35C1*	100.00%	*SLC39A14*	100.00%
SLC39A8	100.00%	*SLC3A1*	100.00%	*SLC4A10*	100.00%
SLC4A4	100.00%	*SLC52A2*	100.00%	*SLC52A3*	100.00%
SLC5A7	100.00%	*SLC6A1*	100.00%	*SLC6A17*	100.00%
SLC6A19	100.00%	*SLC6A3*	100.00%	*SLC6A4*	100.00%
SLC6A5	100.00%	*SLC6A8*	100.00%	*SLC6A9*	100.00%
SLC7A7	100.00%	*SLC9A6*	100.00%	*SLC9A9*	100.00%
SLCO1B3	100.00%	*SMAD4*	100.00%	*SMARCA2*	100.00%
SMARCA4	100.00%	*SMARCB1*	100.00%	*SMARCC2*	100.00%
SMARCE1	100.00%	*SMC1A*	100.00%	*SMC3*	100.00%
SMCHD1	100.00%	*SMPD1*	100.00%	*SMPD4*	100.00%
SMS	100.00%	*SNAI2*	100.00%	*SNAP25*	100.00%
SNAP29	100.00%	*SNCA*	100.00%	*SNCB*	100.00%
SNIP1	100.00%	*SNTA1*	100.00%	*SNX14*	100.00%
SNX27	100.00%	*SOBP*	100.00%	*SOD1*	100.00%
SOD2	100.00%	*SON*	100.00%	*SORL1*	100.00%

SOS1	100.00%	SOX10	100.00%	SOX11	100.00%
SPG7	100.00%	*SPR*	100.00%	*SPTAN1*	100.00%
SPTBN2	100.00%	*SPTBN4*	100.00%	*SPTLC1*	100.00%
SPTLC2	100.00%	*SQSTM1*	100.00%	*SRCAP*	100.00%
SRD5A3	100.00%	*SSR4*	100.00%	*ST3GAL3*	100.00%
ST3GAL5	100.00%	*STAC3*	100.00%	*STAG1*	100.00%
STAMBP	100.00%	*STAR*	100.00%	*STAT1*	100.00%
STAT2	100.00%	*STIL*	100.00%	*STIM1*	100.00%
STRA6	100.00%	*STRADA*	100.00%	*STT3A*	100.00%
STUB1	100.00%	*STX1B*	100.00%	*STXBP1*	100.00%
SUCLA2	100.00%	*SUCLG1*	100.00%	*SUGCT*	100.00%
SUMF1	100.00%	*SUN2*	100.00%	*SUOX*	100.00%
SURF1	100.00%	*SYN1*	100.00%	*SYN2*	100.00%
SYNE1	100.00%	*SYNE2*	100.00%	*SYNGAP1*	100.00%
SYNJ1	100.00%	*SYP*	100.00%	*SYT2*	100.00%
SZT2	100.00%	*TACO1*	100.00%	*TAF1*	100.00%
TAF13	100.00%	*TAF15*	100.00%	*TAF2*	100.00%
TAF6	100.00%	*TAFAZZIN*	100.00%	*TANGO2*	100.00%
TAOK1	100.00%	*TARDBP*	100.00%	*TARS2*	100.00%
TBC1D20	100.00%	*TBC1D23*	100.00%	*TBC1D24*	100.00%
TBCD	100.00%	*TBCE*	100.00%	*TBCK*	100.00%
TBK1	100.00%	*TBL1XR1*	100.00%	*TBR1*	100.00%
TBX1	100.00%	*TBX3*	100.00%	*TCAP*	100.00%
TCF20	100.00%	*TCF4*	100.00%	*TCIRG1*	100.00%
TCOF1	100.00%	*TCTN1*	100.00%	*TCTN2*	100.00%
TCTN3	100.00%	*TDP1*	100.00%	*TDP2*	100.00%
TECPR2	100.00%	*TECR*	100.00%	*TECTA*	100.00%
TENM4	100.00%	*TET2*	100.00%	*TFAP2A*	100.00%
TFAP2B	100.00%	*TFG*	100.00%	*TFR2*	100.00%
TG	100.00%	*TGFB1*	100.00%	*TGFB3*	100.00%
TGIF1	100.00%	*TGM6*	100.00%	*TH*	100.00%
THAP1	100.00%	*THRA*	100.00%	*THRB*	100.00%
TIA1	100.00%	*TIMM50*	100.00%	*TIMM8A*	100.00%
TIMMDC1	100.00%	*TINF2*	100.00%	*TK2*	100.00%

TLK2	100.00%	TMCO1	100.00%	TMEM106B	100.00%
TMEM240	100.00%	*TMEM43*	100.00%	*TMEM67*	100.00%
TMEM70	100.00%	*TMLHE*	100.00%	*TMTC3*	100.00%
TMX2	100.00%	*TNIK*	100.00%	*TNK2*	100.00%
TNNI2	100.00%	*TNNT1*	100.00%	*TNNT3*	100.00%
TNPO3	100.00%	*TOE1*	100.00%	*TOP3A*	100.00%
TOR1A	100.00%	*TOR1AIP1*	100.00%	*TPI1*	100.00%
TPK1	100.00%	*TPM2*	100.00%	*TPM3*	100.00%
TPO	100.00%	*TPP1*	100.00%	*TRAF3IP1*	100.00%
TRAF7	100.00%	*TRAK1*	100.00%	*TRAPPC11*	100.00%
TRAPPC4	100.00%	*TRAPPC9*	100.00%	*TREM2*	100.00%
TREX1	100.00%	*TRIM2*	100.00%	*TRIM32*	100.00%
TRIM8	100.00%	*TRIO*	100.00%	*TRIP12*	100.00%
TRIP4	100.00%	*TRIT1*	100.00%	*TRMT10A*	100.00%
TRMT10C	100.00%	*TRMT5*	100.00%	*TRMU*	100.00%
TRNT1	100.00%	*TRPC6*	100.00%	*TRPM1*	100.00%
TRPM6	100.00%	*TRPS1*	100.00%	*TRPV4*	100.00%
TRRAP	100.00%	*TSC1*	100.00%	*TSC2*	100.00%
TSEN15	100.00%	*TSEN2*	100.00%	*TSEN34*	100.00%
TSEN54	100.00%	*TSFM*	100.00%	*TSHB*	100.00%
TSHR	100.00%	*TSPAN7*	100.00%	*TTBK2*	100.00%
TTC19	100.00%	*TTC21B*	100.00%	*TTC8*	100.00%
TTI2	100.00%	*TTN*	100.00%	*TTPA*	100.00%
TTR	100.00%	*TUBA1A*	100.00%	*TUBA4A*	100.00%
TUBA8	100.00%	*TUBB2A*	100.00%	*TUBB2B*	100.00%
TUBB3	100.00%	*TUBB4A*	100.00%	*TUBG1*	100.00%
TUBGCP4	100.00%	*TUBGCP6*	100.00%	*TUFM*	100.00%
TUSC3	100.00%	*TWIST1*	100.00%	*TWNK*	100.00%
TYMP	100.00%	*TYR*	100.00%	*TYROBP*	100.00%
UBA1	100.00%	*UBA5*	100.00%	*UBAP1*	100.00%
UBE2A	100.00%	*UBE3A*	100.00%	*UBE3B*	100.00%
UBQLN2	100.00%	*UBR1*	100.00%	*UBTF*	100.00%
UCHL1	100.00%	*UFM1*	100.00%	*UGP2*	100.00%
UMPS	100.00%	*UNC80*	100.00%	*UNG*	100.00%

UPB1	100.00%	UPF3B	100.00%	UQCC2	100.00%
VAMP2	100.00%	*VANGL1*	100.00%	*VAPB*	100.00%
VARS1	100.00%	*VARS2*	100.00%	*VCP*	100.00%
VDR	100.00%	*VHL*	100.00%	*VIPAS39*	100.00%
VLDLR	100.00%	*VMA21*	100.00%	*VPS11*	100.00%
VPS13A	100.00%	*VPS13B*	100.00%	*VPS13C*	100.00%
VPS13D	100.00%	*VPS33B*	100.00%	*VPS35*	100.00%
VPS37A	100.00%	*VPS53*	100.00%	*VRK1*	100.00%
WAC	100.00%	*WARS2*	100.00%	*WASF1*	100.00%
WASHC4	100.00%	*WASHC5*	100.00%	*WDFY3*	100.00%
WDR26	100.00%	*WDR37*	100.00%	*WDR45*	100.00%
WDR45B	100.00%	*WDR62*	100.00%	*WDR73*	100.00%
WDR81	100.00%	*WFS1*	100.00%	*WNK1*	100.00%
WNT1	100.00%	*WNT5A*	100.00%	*WNT7A*	100.00%
WWOX	100.00%	*XK*	100.00%	*XPNPEP3*	100.00%
XPR1	100.00%	*YAP1*	100.00%	*YARS1*	100.00%
YARS2	100.00%	*YWHAE*	100.00%	*YWHAG*	100.00%
YY1	100.00%	*ZBTB16*	100.00%	*ZBTB18*	100.00%
ZBTB20	100.00%	*ZBTB24*	100.00%	*ZC3H14*	100.00%
ZC4H2	100.00%	*ZDHHC9*	100.00%	*ZEB2*	100.00%
ZFYVE26	100.00%	*ZFYVE27*	100.00%	*ZIC1*	100.00%
ZIC2	100.00%	*ZIC3*	100.00%	*ZMYND11*	100.00%
ZNF142	100.00%	*ZNF292*	100.00%	*ZNF335*	100.00%
ZNF41	100.00%	*ZNF423*	100.00%	*ZNF699*	100.00%
ZNF711	100.00%	*ZNF81*	100.00%	*CPLANE1*	100.00%
LMNB2	100.00%	*NCDN*	100.00%	*IL6ST*	100.00%
PIK3R1	100.00%	*OTX2*	100.00%	*SNRPN*	100.00%
CDH2	100.00%	*MED27*	100.00%	*FANCF*	100.00%
ERCC4	100.00%	*MYF6*	100.00%	*RSPH1*	100.00%
PHKB	100.00%	*NEFL*	100.00%	*PHF21A*	100.00%
RIMS2	100.00%	*PLAGL1*	100.00%	*MEG3*	100.00%
WIPI2	100.00%	*CENPT*	100.00%	*COLEC10*	100.00%
FANCA	100.00%	*TCF12*	100.00%	*TGFBR1*	100.00%
PIBF1	100.00%	*DHX37*	100.00%	*DDX6*	100.00%

PIK3C2A	100.00%	FAM20C	100.00%	RPS26	100.00%
USF3	100.00%	*OCA2*	100.00%	*ZPR1*	100.00%
SLC44A1	100.00%	*ARNT2*	100.00%	*SPEN*	100.00%
RPL15	100.00%	*RTL1*	100.00%	*SLC12A1*	100.00%
DNAAF11	100.00%	*FAM111A*	100.00%	*TERT*	100.00%
VAX1	100.00%	*NDN*	100.00%	*LSM11*	100.00%
HPDL	100.00%	*CDK19*	100.00%	*SMARCD1*	100.00%
DCPS	100.00%	*GLIS3*	100.00%	*CCDC65*	100.00%
EMG1	100.00%	*TERC*	100.00%	*GTF2E2*	100.00%
FANCE	100.00%	*SYNCRIP*	100.00%	*DBR1*	100.00%
SFTPC	100.00%	*AGO2*	100.00%	*RPS19*	100.00%
HACD1	100.00%	*TRIP13*	100.00%	*SPTBN1*	100.00%
CHST3	100.00%	*TELO2*	100.00%	*USP27X*	100.00%
PLXND1	100.00%	*IARS1*	100.00%	*CDK8*	100.00%
BRCA1	100.00%	*POLR3GL*	100.00%	*IQCB1*	100.00%
SUPT16H	100.00%	*DRC1*	100.00%	*ATG5*	100.00%
STAG2	100.00%	*MEGF8*	100.00%	*ZMIZ1*	100.00%
NKX2-5	100.00%	*PCK1*	100.00%	*SEMA3E*	100.00%
RPL5	100.00%	*SCUBE3*	100.00%	*UFD1*	100.00%
EN1	100.00%	*COL3A1*	100.00%	*KCNAB2*	100.00%
PPP1R15B	100.00%	*PITRM1*	100.00%	*NUP107*	100.00%
DSE	100.00%	*TANC2*	100.00%	*UBR7*	100.00%
SRP54	100.00%	*TNFRSF11B*	100.00%	*ZNF592*	100.00%
USH1C	100.00%	*HS2ST1*	100.00%	*RSRC1*	100.00%
DLL1	100.00%	*GDF6*	100.00%	*TMEM107*	100.00%
BPTF	100.00%	*MRPS25*	100.00%	*PGAP3*	100.00%
CNOT3	100.00%	*TUBB*	100.00%	*HNRNPK*	100.00%
SLC12A2	100.00%	*ERMARD*	100.00%	*DNAI2*	100.00%
RNU4ATAC	100.00%	*SEC23A*	100.00%	*HYLS1*	100.00%
TRAPPC12	100.00%	*PRUNE1*	100.00%	*MED12L*	100.00%
ARL3	100.00%	*POU4F1*	100.00%	*ZMYM2*	100.00%
FUZ	100.00%	*ARF1*	100.00%	*ITGA2*	100.00%
RFWD3	100.00%	*FITM2*	100.00%	*MDH1*	100.00%
DDX11	100.00%	*DLK1*	100.00%	*MID2*	100.00%

CWC27	100.00%	YIPF5	100.00%	CDC6	100.00%
PTCH2	100.00%	*DNAH1*	100.00%	*TDGF1*	100.00%
DICER1	100.00%	*WDR35*	100.00%	*BUB1B*	100.00%
LARP7	100.00%	*CFAP300*	100.00%	*MYOD1*	100.00%
IYD	100.00%	*MKRN3*	100.00%	*TASP1*	100.00%
HARS1	100.00%	*PRDM16*	100.00%	*HOXB1*	100.00%
KCNJ11	100.00%	*FZD4*	100.00%	*HOXA2*	100.00%
SLC45A1	100.00%	*FCSK*	100.00%	*CCDC174*	100.00%
KYNU	100.00%	*LNPK*	100.00%	*MICOS13*	100.00%
LRP5	100.00%	*SIM1*	100.00%	*PRR12*	100.00%
DNAAF4	100.00%	*KIAA0753*	100.00%	*HMGA2*	100.00%
CLDN11	100.00%	*NTNG2*	100.00%	*FARSA*	100.00%
CEP120	100.00%	*ZBTB11*	100.00%	*NONO*	100.00%
GRM7	100.00%	*CLTRN*	100.00%	*NODAL*	100.00%
VPS41	100.00%	*LAGE3*	100.00%	*SPOP*	100.00%
PLOD3	100.00%	*PPP1CB*	100.00%	*PIK3CD*	100.00%
PTH1R	100.00%	*NARS1*	100.00%	*SATB1*	100.00%
BRCA2	100.00%	*TBX2*	100.00%	*WDR11*	100.00%
NXN	100.00%	*CDT1*	100.00%	*GRHL3*	100.00%
BMP4	100.00%	*PLOD1*	100.00%	*LCA5*	100.00%
TKT	100.00%	*CNOT1*	100.00%	*MCTP2*	100.00%
SRPX2	100.00%	*FBXW11*	100.00%	*RNU12*	100.00%
CAMKMT	100.00%	*VPS33A*	100.00%	*CFAP298*	100.00%
PGM3	100.00%	*SARS1*	100.00%	*RBPJ*	100.00%
IGF2	100.00%	*DNAAF3*	100.00%	*ARVCF*	100.00%
FOXA2	100.00%	*SPEF2*	100.00%	*FZD2*	100.00%
GUCY2D	100.00%	*DNAI1*	100.00%	*BCAS3*	100.00%
FAR1	100.00%	*SAMD9*	100.00%	*RPS20*	100.00%
PAX8	100.00%	*NIN*	100.00%	*PLCB4*	100.00%
GTF2H5	100.00%	*B3GALT6*	100.00%	*HS6ST2*	100.00%
ACD	100.00%	*LAS1L*	100.00%	*COL27A1*	100.00%
CHN1	100.00%	*LRRK1*	100.00%	*MVK*	100.00%
FGFRL1	100.00%	*HTT*	100.00%	*ANK1*	100.00%
SPECC1L	100.00%	*CCDC47*	100.00%	*BRIP1*	100.00%

AQP2	100.00%	EXOC8	100.00%	SOX6	100.00%
UFC1	100.00%	*HELLS*	100.00%	*LETM1*	100.00%
CCDC103	100.00%	*SPINK5*	100.00%	*JAG2*	100.00%
CARS1	100.00%	*MYO5B*	100.00%	*PPP1R21*	100.00%
VAC14	100.00%	*ATP5F1D*	100.00%	*DNAH11*	100.00%
TMEM218	100.00%	*WDR4*	100.00%	*NEMF*	100.00%
AP1G1	100.00%	*JARID2*	100.00%	*SLC9A1*	100.00%
CCDC28B	100.00%	*SMG8*	100.00%	*RTEL1*	100.00%
CKAP2L	100.00%	*MNX1*	100.00%	*PCNA*	100.00%
PNP	100.00%	*UBB*	100.00%	*CLCN7*	100.00%
RSPRY1	100.00%	*HAL*	100.00%	*ACOX2*	100.00%
FGFR1	100.00%	*GORAB*	100.00%	*COL1A2*	100.00%
CRKL	100.00%	*SLC18A2*	100.00%	*RNU7-1*	100.00%
TBX4	100.00%	*B4GALT7*	100.00%	*POU3F3*	100.00%
SHANK3	100.00%	*KNSTRN*	100.00%	*RSPH9*	100.00%
FGF3	100.00%	*MMP13*	100.00%	*TTC37*	100.00%
COPB2	100.00%	*MRPL12*	100.00%	*IRF6*	100.00%
BCORL1	100.00%	*DNAJB13*	100.00%	*NPAP1*	100.00%
SYT14	100.00%	*NKX2-1*	100.00%	*STS*	100.00%
CADM3	100.00%	*DNAH5*	100.00%	*IL37*	100.00%
USP45	100.00%	*PRKD1*	100.00%	*NDUFA13*	100.00%
VPS4A	100.00%	*TENT5A*	100.00%	*SPAG1*	100.00%
DNMT3B	100.00%	*TPP2*	100.00%	*ITCH*	100.00%
ANAPC1	100.00%	*BCL11B*	100.00%	*TSPAN12*	100.00%
SLC26A4	100.00%	*PTDSS1*	100.00%	*LRRC32*	100.00%
CDH23	100.00%	*ZNFX1*	100.00%	*XRCC2*	100.00%
EXOSC2	100.00%	*TNFRSF11A*	100.00%	*ZMYND10*	100.00%
ORC4	100.00%	*NOVA2*	100.00%	*DONSON*	100.00%
TALDO1	100.00%	*INSR*	100.00%	*TRMT1*	100.00%
RPS28	100.00%	*DCHS1*	100.00%	*RPL35*	100.00%
ANTXR1	100.00%	*TNPO2*	100.00%	*FRMD4A*	100.00%
EDEM3	100.00%	*LSS*	100.00%	*PARN*	100.00%
AIPL1	100.00%	*SLC37A4*	100.00%	*HIRA*	100.00%
MCIDAS	100.00%	*METTL5*	100.00%	*PALB2*	100.00%

FANCG	100.00%	ACBD5	100.00%	SLC39A13	100.00%
ANKRD17	100.00%	*CDKN1C*	100.00%	*KIAA0586*	100.00%
CELF2	100.00%	*USP7*	100.00%	*FOXF1*	100.00%
CCDC39	100.00%	*CCNO*	100.00%	*KLC2*	100.00%
TRIP11	100.00%	*FOXJ1*	100.00%	*SLX4*	100.00%
MORC2	100.00%	*RPGR*	100.00%	*PIGY*	100.00%
ASXL2	100.00%	*SMAD6*	100.00%	*FMR1*	100.00%
RNF2	100.00%	*CDCA7*	100.00%	*INPP5K*	100.00%
EFL1	100.00%	*LIG4*	100.00%	*PDE4D*	100.00%
LTBP4	100.00%	*LTC4S*	100.00%	*SUFU*	100.00%
HEPHL1	100.00%	*ALKBH8*	100.00%	*ACAT2*	100.00%
GEMIN4	100.00%	*KCNA4*	100.00%	*CTNND2*	100.00%
GPSM2	100.00%	*AK9*	100.00%	*SLC5A5*	100.00%
EXTL3	100.00%	*GJB6*	100.00%	*ORC6*	100.00%
CLIC2	100.00%	*GATA1*	100.00%	*RAD51*	100.00%
DNAAF2	100.00%	*MAPK8IP3*	100.00%	*RPS23*	100.00%
PCYT1A	100.00%	*AMER1*	100.00%	*CTBP1*	100.00%
LRRC56	100.00%	*TPRKB*	100.00%	*RDH12*	100.00%
RRAS2	100.00%	*NFKB2*	100.00%	*DDR2*	100.00%
ARMC9	100.00%	*EDN1*	100.00%	*ATG7*	100.00%
MADD	100.00%	*DPYSL5*	100.00%	*ERF*	100.00%
PHKG2	100.00%	*PLAG1*	100.00%	*EIF2AK2*	100.00%
IQSEC1	100.00%	*OSTM1*	100.00%	*SLC10A7*	100.00%
STX11	100.00%	*SMG9*	100.00%	*TOPORS*	100.00%
TSPOAP1	100.00%	*CYP3A4*	100.00%	*IFT52*	100.00%
ATXN7	100.00%	*DIS3L2*	100.00%	*DVL1*	100.00%
CDK13	100.00%	*FANCI*	100.00%	*SPATA7*	100.00%
PROC	100.00%	*EFEMP2*	100.00%	*CLIP1*	100.00%
SLC46A1	100.00%	*PRDX1*	100.00%	*TNRC6B*	100.00%
NAA20	100.00%	*COQ5*	100.00%	*FBXO31*	100.00%
REV3L	100.00%	*FLNB*	100.00%	*ADARB1*	100.00%
PTPRQ	100.00%	*FLII*	100.00%	*COL1A1*	100.00%
GAS2L2	100.00%	*SETD1B*	100.00%	*ZFHX4*	100.00%
PPIL1	100.00%	*RLIM*	100.00%	*H19*	100.00%

SH2B1	100.00%	GP1BB	100.00%	AMN	100.00%
SIX6	100.00%	*SGMS2*	100.00%	*PPARG*	100.00%
SLC9A7	100.00%	*CDC42*	100.00%	*RNF125*	100.00%
GEMIN5	100.00%	*EYA1*	100.00%	*CLCN6*	100.00%
GPR161	100.00%	*PROKR2*	100.00%	*ZNF407*	100.00%
GNB2	100.00%	*BUB3*	100.00%	*RIC1*	100.00%
ROBO1	100.00%	*FOXP3*	100.00%	*DNAL1*	100.00%
ACP5	100.00%	*P3H1*	100.00%	*CYP2R1*	100.00%
ZSWIM6	100.00%	*MIR17HG*	100.00%	*RRP7A*	100.00%
DNAH9	100.00%	*LMBRD2*	100.00%	*RPL26*	100.00%
GATA4	100.00%	*COX8A*	100.00%	*YME1L1*	100.00%
KCNJ5	100.00%	*CRB1*	100.00%	*BMPER*	100.00%
IPO8	100.00%	*CSNK2A1*	100.00%	*SF3B4*	100.00%
NDUFC2	100.00%	*CAPN15*	100.00%	*DHX16*	100.00%
UQCRFS1	100.00%	*GATA6*	100.00%	*RAC3*	100.00%
TP53	100.00%	*UNC45A*	100.00%	*ERCC3*	100.00%
EXOSC5	100.00%	*CRAT*	100.00%	*FANCD2*	100.00%
ODC1	100.00%	*PYGL*	100.00%	*STT3B*	100.00%
TOGARAM1	100.00%	*LTBP1*	100.00%	*GMNN*	100.00%
TXN2	100.00%	*MAD2L2*	100.00%	*IL1RN*	100.00%
TMEM94	100.00%	*REPS1*	100.00%	*KCNN3*	100.00%
GALNT2	100.00%	*TGFBR2*	100.00%	*ALMS1*	100.00%
DYNC1I2	100.00%	*LEMD3*	100.00%	*FLRT1*	100.00%
ODAD1	100.00%	*NUP214*	100.00%	*TENM3*	100.00%
SIK3	100.00%	*CEP104*	100.00%	*FGF8*	100.00%
CSGALNACT1	100.00%	*GAS1*	100.00%	*KLLN*	100.00%
NLGN1	100.00%	*PUS7*	100.00%	*SPATA5L1*	100.00%
WDR19	100.00%	*FLI1*	100.00%	*SIAH1*	100.00%
KCNJ13	100.00%	*MYO7A*	100.00%	*EMC1*	100.00%
DNAJC21	100.00%	*GNAI3*	100.00%	*KDM3B*	100.00%
MAPRE2	100.00%	*CCND2*	100.00%	*AFF4*	100.00%
PIGK	100.00%	*KDM6B*	100.00%	*YIF1B*	100.00%
CPE	100.00%	*FGF13*	100.00%	*CDK5*	100.00%
PSMG2	100.00%	*ZNF148*	100.00%	*UGT1A1*	100.00%

IMPDH1	100.00%	GHR	100.00%	CENPE	100.00%
UGDH	100.00%	*DNAAF5*	100.00%	*NIPA2*	100.00%
NOP10	100.00%	*DUOX2*	100.00%	*TRHR*	100.00%
UBE2T	100.00%	*INTS1*	100.00%	*NDUFA8*	100.00%
RPL31	100.00%	*TRAIP*	100.00%	*ESPN*	100.00%
TSR2	100.00%	*TRAPPC6B*	100.00%	*BTK*	100.00%
KLHL15	100.00%	*DALRD3*	100.00%	*FLCN*	100.00%
ZNHIT3	100.00%	*FIBP*	100.00%	*ALB*	100.00%
DLG1	100.00%	*ADA2*	100.00%	*SALL4*	100.00%
COLEC11	100.00%	*COL5A2*	100.00%	*NTNG1*	100.00%
TRAPPC2L	100.00%	*NPHP4*	100.00%	*ABL1*	100.00%
RPS10	100.00%	*ATP10A*	100.00%	*HNF1B*	100.00%
FANCL	100.00%	*POU3F4*	100.00%	*CCDC32*	100.00%
ZNF408	100.00%	*PDE2A*	100.00%	*UQCC3*	100.00%
PIGF	100.00%	*COG2*	100.00%	*POLR2A*	100.00%
KCNJ6	100.00%	*PIEZO1*	100.00%	*SMARCD2*	100.00%
STEEP1	100.00%	*SLC2A2*	100.00%	*ADD3*	100.00%
PHEX	100.00%	*PI4KA*	100.00%	*RAD51C*	100.00%
CDK10	100.00%	*BMPR1A*	100.00%	*RPS24*	100.00%
MYSM1	100.00%	*SMO*	100.00%	*RPGRIP1*	100.00%
ZNF462	100.00%	*TMEM222*	100.00%	*COL25A1*	100.00%
NAT8L	100.00%	*AEBP1*	100.00%	*SARDH*	100.00%
XRCC4	100.00%	*TTC12*	100.00%	*PRKAR1A*	100.00%
PUF60	100.00%	*XYLT1*	100.00%	*GAS8*	100.00%
UBE4B	100.00%	*ARCN1*	100.00%	*NDUFAF8*	100.00%
NHLRC2	100.00%	*IDH1*	100.00%	*AP3D1*	100.00%
IFT74	100.00%	*SOX4*	100.00%	*SLC5A6*	100.00%
PAX1	100.00%	*SP7*	100.00%	*MMP23B*	100.00%
HAAO	100.00%	*DNAAF1*	100.00%	*PEPD*	100.00%
CCNK	100.00%	*DLX4*	100.00%	*ATRIP*	100.00%
AVPR2	100.00%	*CUBN*	100.00%	*SEC24D*	100.00%
ZFR	100.00%	*BCR*	100.00%	*NKAP*	100.00%
LUZP1	100.00%	*COL5A1*	100.00%	*SHMT2*	100.00%
SOS2	100.00%	*FTCD*	100.00%	*ELN*	100.00%

DCC	100.00%	EED	100.00%	FANCC	100.00%
SDHC	100.00%	*COPB1*	100.00%	*RPL27*	100.00%
TKFC	100.00%	*NDUFB10*	100.00%	*CD109*	100.00%
TMEM63A	100.00%	*AKT1*	100.00%	*PPP1R12A*	100.00%
CEP57	100.00%	*NDUFA4*	100.00%	*ITGA2B*	100.00%
SPARC	100.00%	*GALE*	100.00%	*SNORD118*	100.00%
BMP1	100.00%	*OTUD5*	100.00%	*MSL3*	100.00%
TCF7L2	100.00%	*CLTCL1*	100.00%	*POC1A*	100.00%
FDFT1	100.00%	*IPW*	100.00%	*TNR*	100.00%
TFE3	100.00%	*EIF2AK3*	100.00%	*GON7*	100.00%
SNORD115-1	100.00%	*MESD*	100.00%	*THOC6*	100.00%
SVBP	100.00%	*CANT1*	100.00%	*CEP85L*	100.00%
PET117	100.00%	*RFC1*	100.00%	*PRKCZ*	100.00%
VPS51	100.00%	*THG1L*	100.00%	*TPRN*	100.00%
HCN4	100.00%	*EXOC7*	100.00%	*DYNC2I2*	100.00%
LINGO1	100.00%	*SCN11A*	100.00%	*STX3*	100.00%
ARHGAP29	100.00%	*HERC1*	100.00%	*PDPN*	100.00%
GJB2	100.00%	*GGPS1*	100.00%	*SEC24C*	100.00%
NHP2	100.00%	*EZR*	100.00%	*APC2*	100.00%
EXOSC1	100.00%	*G6PC3*	100.00%	*ACTN2*	100.00%
GSX2	100.00%	*NUP85*	100.00%	*NMNAT1*	100.00%
MRPS14	100.00%	*LHX1*	100.00%	*MIR140*	100.00%
MRAS	100.00%	*ENPP1*	100.00%	*ARHGEF2*	100.00%
FOS	100.00%	*PCGF2*	100.00%	*TMEM185A*	100.00%
USH1G	100.00%	*RPS29*	100.00%	*RREB1*	100.00%
MN1	100.00%	*SH3PXD2B*	100.00%	*OGT*	100.00%
MYO1H	100.00%	*SKIV2L*	100.00%	*KIF4A*	100.00%
INS	100.00%	*KDM1A*	100.00%	*NELFA*	100.00%
RPL18	100.00%	*MAP3K20*	100.00%	*FOXE1*	100.00%
GCLC	100.00%	*SGPL1*	100.00%	*GPR88*	100.00%
TBC1D7	100.00%	*NEPRO*	100.00%	*POR*	100.00%
PPM1B	100.00%	*HMGB3*	100.00%	*FBLN1*	100.00%
MAP1B	100.00%	*ATP6V1B2*	100.00%	*FBXL3*	100.00%
COA3	100.00%	*USB1*	100.00%	*BICRA*	100.00%

CRX	100.00%	CDC40	100.00%	TULP1	100.00%
ODAD2	100.00%	*WBP11*	100.00%	*SNORD116-1*	100.00%
CFAP221	100.00%	*C2orf69*	100.00%	*KAT5*	100.00%
MRM2	100.00%	*VPS35L*	100.00%	*CNOT2*	100.00%
RPS17	100.00%	*INTS8*	100.00%	*BMP2*	100.00%
CDK6	100.00%	*TMEM38B*	100.00%	*RPS27*	100.00%
INTU	100.00%	*RSPH3*	100.00%	*PALS1*	100.00%
TET3	100.00%	*MAST1*	100.00%	*KIF15*	100.00%
TTC5	100.00%	*SPRED1*	100.00%	*LARS1*	100.00%
ZFP57	100.00%	*PWAR1*	100.00%	*MAFB*	100.00%
EXOC2	100.00%	*FANCM*	100.00%	*FAT4*	100.00%
THOC2	100.00%	*STK36*	100.00%	*DPH1*	100.00%
VPS45	100.00%	*NCAPG2*	100.00%	*RD3*	100.00%
B3GAT3	100.00%	*UFSP2*	100.00%	*GYS2*	100.00%
GP1BA	100.00%	*NRROS*	100.00%	*CASZ1*	100.00%
ADAMTS2	100.00%	*POLR1B*	100.00%	*ATP6V1E1*	100.00%
IKBKG	100.00%	*PWRN1*	100.00%	*CFAP43*	100.00%
PHC1	100.00%	*WRAP53*	100.00%	*MCM3AP*	100.00%
TP63	100.00%	*NME8*	100.00%	*CDH1*	100.00%
ARHGEF28	100.00%	*CIZ1*	100.00%	*DAO*	100.00%
NR4A2	100.00%	*SS18L1*	100.00%	*AR*	100.00%
ARSI	100.00%	*ATXN1*	100.00%	*ATXN10*	100.00%
ATXN2	100.00%	*BEAN1*	100.00%	*C9orf72*	100.00%
CDK16	100.00%	*DRD2*	100.00%	*DRD5*	100.00%
HFE	100.00%	*IPPK*	100.00%	*JPH3*	100.00%
KLC4	100.00%	*MAT1A*	100.00%	*MR1*	100.00%
MT-ATP6	100.00%	*MT-ND6*	100.00%	*NOP56*	100.00%
PAX2	100.00%	*PPP2R2B*	100.00%	*SAR1B*	100.00%
SCP2	100.00%	*SLC41A1*	100.00%	*SLC52A1*	100.00%
SNCAIP	100.00%	*TBP*	100.00%	*UBR4*	100.00%
UNC13A	100.00%	*VEGFA*	100.00%	*WDR48*	100.00%
XRCC1	100.00%	*DNAJC7*	100.00%	*GLT8D1*	100.00%
UQCRC1	100.00%	*DSCAM*	100.00%	*KATNAL2*	100.00%
BAZ2B	100.00%	*CACNA2D3*	100.00%	*DIP2C*	100.00%

ERBIN	100.00%	ILF2	100.00%	INTS6	100.00%
USP15	100.00%	*ABCA10*	100.00%	*ABCA13*	100.00%
ADCY3	100.00%	*ADORA2A*	100.00%	*ADORA3*	100.00%
ADRB2	100.00%	*AGAP1*	100.00%	*AGAP2*	100.00%
AGBL4	100.00%	*AGMO*	100.00%	*AGO1*	100.00%
AKAP9	100.00%	*ANKS1B*	100.00%	*ANXA1*	100.00%
APBA2	100.00%	*APBB1*	100.00%	*APH1A*	100.00%
ARHGAP11B	100.00%	*ARHGAP32*	100.00%	*ARHGAP5*	100.00%
ASAP2	100.00%	*ASMT*	100.00%	*ATP2B2*	100.00%
AVPR1A	100.00%	*AVPR1B*	100.00%	*AZGP1*	100.00%
BCAS1	100.00%	*BICDL1*	100.00%	*BIRC6*	100.00%
BRD4	100.00%	*BST1*	100.00%	*BTAF1*	100.00%
C15orf62	100.00%	*C4B*	100.00%	*CA6*	100.00%
CACNA1I	100.00%	*CACNA2D1*	100.00%	*CADM1*	100.00%
CADM2	100.00%	*CADPS2*	100.00%	*CAMK4*	100.00%
CAPN12	100.00%	*CAPRIN1*	100.00%	*CCDC91*	100.00%
CCNG1	100.00%	*CCT4*	100.00%	*CD276*	100.00%
CD38	100.00%	*CD99L2*	100.00%	*CDC42BPB*	100.00%
CDH10	100.00%	*CDH13*	100.00%	*CDH22*	100.00%
CDH8	100.00%	*CDH9*	100.00%	*CECR2*	100.00%
CELF4	100.00%	*CELF6*	100.00%	*CGNL1*	100.00%
CHRM3	100.00%	*CHRNB3*	100.00%	*CLASP1*	100.00%
CMIP	100.00%	*CNR1*	100.00%	*CNTN3*	100.00%
CNTN5	100.00%	*CNTN6*	100.00%	*CNTNAP3*	100.00%
CNTNAP4	100.00%	*CNTNAP5*	100.00%	*COL28A1*	100.00%
CPEB4	100.00%	*CSNK1E*	100.00%	*CTTNBP2*	100.00%
CYFIP1	100.00%	*CYLC2*	100.00%	*DAGLA*	100.00%
DAPP1	100.00%	*DDX53*	100.00%	*DENR*	100.00%
DIP2A	100.00%	*DISC1*	100.00%	*DIXDC1*	100.00%
DLGAP1	100.00%	*DLGAP3*	100.00%	*DLX2*	100.00%
DLX6	100.00%	*DNAH10*	100.00%	*DNAH17*	100.00%
DNAH3	100.00%	*DNER*	100.00%	*DOCK1*	100.00%
DOCK4	100.00%	*DPP10*	100.00%	*DPP4*	100.00%
DPYSL2	100.00%	*DPYSL3*	100.00%	*DRD1*	100.00%

DUSP15	100.00%	DYDC1	100.00%	DYDC2	100.00%
EP400	100.00%	*EPC2*	100.00%	*EPHB2*	100.00%
EPPK1	100.00%	*ERG*	100.00%	*ERMN*	100.00%
ESR2	100.00%	*ESRRB*	100.00%	*EXOC3*	100.00%
EXOC5	100.00%	*EXOC6*	100.00%	*FABP5*	100.00%
FAM47A	100.00%	*FAT1*	100.00%	*FBXO33*	100.00%
FBXO40	100.00%	*FCRL6*	100.00%	*FEZF2*	100.00%
FHIT	100.00%	*FRK*	100.00%	*GABRA4*	100.00%
GABRG3	100.00%	*GALNT13*	100.00%	*GALNT14*	100.00%
GALNT8	100.00%	*GAS2*	100.00%	*GDA*	100.00%
GGNBP2	100.00%	*GIGYF1*	100.00%	*GLIS1*	100.00%
GLRA2	100.00%	*GNB1L*	100.00%	*GPD2*	100.00%
GPR37	100.00%	*GPR85*	100.00%	*GRID1*	100.00%
GRID2IP	100.00%	*GRIK3*	100.00%	*GRIK4*	100.00%
GRIK5	100.00%	*GRM5*	100.00%	*GSTM1*	100.00%
GTF2I	100.00%	*GUCY1A2*	100.00%	*HDLBP*	100.00%
HECTD4	100.00%	*HIVEP3*	100.00%	*HMGN1*	100.00%
HOMER1	100.00%	*HS3ST5*	100.00%	*HTR1B*	100.00%
HTR3A	100.00%	*HTR3C*	100.00%	*ICA1*	100.00%
IL1R2	100.00%	*IL1RAPL2*	100.00%	*IMMP2L*	100.00%
INPP1	100.00%	*IQGAP3*	100.00%	*KATNAL1*	100.00%
KCND2	100.00%	*KCNJ15*	100.00%	*KCNK7*	100.00%
KCTD13	100.00%	*KDM4C*	100.00%	*KHDRBS2*	100.00%
KIAA1586	100.00%	*KIF13B*	100.00%	*KLF16*	100.00%
KRR1	100.00%	*KRT26*	100.00%	*LILRB2*	100.00%
LIN7B	100.00%	*LRFN2*	100.00%	*LRFN5*	100.00%
LRRC1	100.00%	*LRRC4*	100.00%	*LZTS2*	100.00%
MACROD2	100.00%	*MAOB*	100.00%	*MAPK3*	100.00%
MARK1	100.00%	*MBD1*	100.00%	*MBD3*	100.00%
MBD4	100.00%	*MBD6*	100.00%	*MDGA2*	100.00%
MEGF11	100.00%	*MIR137*	100.00%	*MNT*	100.00%
MSANTD2	100.00%	*MTF1*	100.00%	*MYH10*	100.00%
MYH4	100.00%	*MYO16*	100.00%	*MYO5C*	100.00%
NAALADL2	100.00%	*NAV2*	100.00%	*NCKAP5*	100.00%

NCOR1	100.00%	NDUFA5	100.00%	NEO1	100.00%
NTRK3	100.00%	*NUAK1*	100.00%	*NUDCD2*	100.00%
NXPH1	100.00%	*ODF3L2*	100.00%	*OR1C1*	100.00%
OR2M4	100.00%	*OR2T10*	100.00%	*OR52M1*	100.00%
OTUD7A	100.00%	*OTX1*	100.00%	*OXT*	100.00%
OXTR	100.00%	*P2RX5*	100.00%	*PAK2*	100.00%
PARD3B	100.00%	*PATJ*	100.00%	*PAX5*	100.00%
PCDH10	100.00%	*PCDH11X*	100.00%	*PCDH9*	100.00%
PCDHA1	100.00%	*PCDHA10*	100.00%	*PCDHA11*	100.00%
PCDHA12	100.00%	*PCDHA13*	100.00%	*PCDHA2*	100.00%
PCDHA3	100.00%	*PCDHA4*	100.00%	*PCDHA5*	100.00%
PCDHA6	100.00%	*PCDHA7*	100.00%	*PCDHA8*	100.00%
PCDHA9	100.00%	*PCDHAC1*	100.00%	*PCDHAC2*	100.00%
PDE1C	100.00%	*PER1*	100.00%	*PHF2*	100.00%
PHRF1	100.00%	*PIK3CG*	100.00%	*PLAUR*	100.00%
PLXNA3	100.00%	*PLXNA4*	100.00%	*PLXNB1*	100.00%
PNPLA7	100.00%	*POLA2*	100.00%	*PPFIA1*	100.00%
PPP1R1B	100.00%	*PPP2R1B*	100.00%	*PREX1*	100.00%
PRKCB	100.00%	*PRKDC*	100.00%	*PRPF39*	100.00%
PRUNE2	100.00%	*PSD3*	100.00%	*PTBP2*	100.00%
PTGS2	100.00%	*PTK7*	100.00%	*PTPRB*	100.00%
PTPRT	100.00%	*PYHIN1*	100.00%	*RAB11FIP5*	100.00%
RAB2A	100.00%	*RAB43*	100.00%	*RAD21L1*	100.00%
RAPGEF4	100.00%	*RASSF5*	100.00%	*RBM27*	100.00%
REEP3	100.00%	*RFX3*	100.00%	*RGS7*	100.00%
RHOXF1	100.00%	*RIMS3*	100.00%	*RIT2*	100.00%
RNF38	100.00%	*RPS6KA2*	100.00%	*SAE1*	100.00%
SAMD11	100.00%	*SCFD2*	100.00%	*SDC2*	100.00%
SERPINE1	100.00%	*SETDB1*	100.00%	*SETDB2*	100.00%
SEZ6L2	100.00%	*SGSM3*	100.00%	*SHANK1*	100.00%
SHOX	100.00%	*SLC22A15*	100.00%	*SLC22A9*	100.00%
SLC24A2	100.00%	*SLC25A27*	100.00%	*SLC25A39*	100.00%
SLC29A4	100.00%	*SLC35B1*	100.00%	*SLC38A10*	100.00%
SLC7A3	100.00%	*SLC7A5*	100.00%	*SLITRK5*	100.00%

SMG6	100.00%	SND1	100.00%	SNTG2	100.00%
STYK1	100.00%	*SYAP1*	100.00%	*SYT17*	100.00%
TAF1C	100.00%	*TBC1D31*	100.00%	*TBC1D5*	100.00%
TBL1X	100.00%	*TDO2*	100.00%	*TERB2*	100.00%
TERF2	100.00%	*THBS1*	100.00%	*TM4SF19*	100.00%
TMPRSS9	100.00%	*TOP3B*	100.00%	*TRIM33*	100.00%
TSHZ3	100.00%	*TSPAN17*	100.00%	*TUBGCP5*	100.00%
UBE2H	100.00%	*UBE3C*	100.00%	*UBR5*	100.00%
UNC79	100.00%	*VASH1*	100.00%	*VIL1*	100.00%
VSIG4	100.00%	*WNK3*	100.00%	*XPO1*	100.00%
YEATS2	100.00%	*YTHDC2*	100.00%	*ZC3H4*	100.00%
ZNF18	100.00%	*ZNF385B*	100.00%	*ZNF517*	100.00%
ZNF548	100.00%	*ZNF559*	100.00%	*ZNF626*	100.00%
ZNF713	100.00%	*ZNF774*	100.00%	*ZNF804A*	100.00%
ZNF827	100.00%	*ZWILCH*	100.00%	*ABCC9*	100.00%
ANKH	100.00%	*BLM*	100.00%	*BRSK2*	100.00%
CCBE1	100.00%	*CEP83*	100.00%	*CHD4*	100.00%
CRB2	100.00%	*CSDE1*	100.00%	*DDX59*	100.00%
ERCC6L2	100.00%	*GNAI1*	100.00%	*MAB21L1*	100.00%
MAB21L2	100.00%	*NANS*	100.00%	*RAB23*	100.00%
RARB	100.00%	*RIT1*	100.00%	*SCAPER*	100.00%
SMOC1	100.00%	*SNRPB*	100.00%	*SOX9*	100.00%
TAT	100.00%	*TCN2*	100.00%	*WDPCP*	100.00%
ACVR1	100.00%	*AIRE*	100.00%	*AKR1D1*	100.00%
AP1B1	100.00%	*AP2S1*	100.00%	*ARHGAP35*	100.00%
ARL14EP	100.00%	*ASCC3*	100.00%	*ASTN1*	100.00%
ATP6V0A1	100.00%	*ATXN2L*	100.00%	*CAPZA2*	100.00%
CCDC186	100.00%	*CEP55*	100.00%	*CHD5*	100.00%
CSNK1G1	100.00%	*CTNND1*	100.00%	*CTU2*	100.00%
DDB1	100.00%	*DDX23*	100.00%	*DHX32*	100.00%
DLG2	100.00%	*DPH2*	100.00%	*EEF1B2*	100.00%
EIF5A	100.00%	*EPHA7*	100.00%	*ERGIC3*	100.00%
FAAH2	100.00%	*FAM120C*	100.00%	*FAM50A*	100.00%
FBRSL1	100.00%	*FOXP4*	100.00%	*FRAS1*	100.00%

FREM2	100.00%	FRY	100.00%	GNAI2	100.00%
KLF7	100.00%	*MAPK10*	100.00%	*MAPKAPK5*	100.00%
MMGT1	100.00%	*MSL2*	100.00%	*NBN*	100.00%
NT5C3A	100.00%	*NUDT2*	100.00%	*NUP188*	100.00%
NYX	100.00%	*OXR1*	100.00%	*PCDHGC4*	100.00%
PDCD6IP	100.00%	*PGM2L1*	100.00%	*PJA1*	100.00%
PRKACB	100.00%	*PRKAR1B*	100.00%	*PSMB8*	100.00%
PSMC5	100.00%	*PTHLH*	100.00%	*PTPN4*	100.00%
PTRHD1	100.00%	*RAB14*	100.00%	*RAP1B*	100.00%
RAX	100.00%	*RHEB*	100.00%	*RMRP*	100.00%
SCAF4	100.00%	*SCAMP5*	100.00%	*SEC31A*	100.00%
SIN3B	100.00%	*SLC26A2*	100.00%	*SLC35D1*	100.00%
SLC4A1	100.00%	*SLC4A11*	100.00%	*SMAD3*	100.00%
SMARCA5	100.00%	*SRRM2*	100.00%	*TBC1D2B*	100.00%
TOMM70	100.00%	*TRPM3*	100.00%	*TUBGCP2*	100.00%
U2AF2	100.00%	*UPF1*	100.00%	*XPA*	100.00%
ZNF526	100.00%	*A2ML1*	100.00%	*ABCB11*	100.00%
ABCG5	100.00%	*ACAN*	100.00%	*ACBD6*	100.00%
ACE2	100.00%	*ACIN1*	100.00%	*ACOT9*	100.00%
ADGRG4	100.00%	*ADRA2B*	100.00%	*AFP*	100.00%
AGT	100.00%	*AK1*	100.00%	*AKAP17A*	100.00%
AKAP4	100.00%	*AKAP6*	100.00%	*AKR1C2*	100.00%
ALDH1A3	100.00%	*AQP7*	100.00%	*ARHGAP36*	100.00%
ARHGAP6	100.00%	*ARHGEF4*	100.00%	*ARIH1*	100.00%
ARSF	100.00%	*ASB12*	100.00%	*ASMTL*	100.00%
ASPH	100.00%	*ATP2C2*	100.00%	*ATP6V1B1*	100.00%
ATP8B1	100.00%	*ATXN3L*	100.00%	*AVP*	100.00%
AWAT2	100.00%	*BDP1*	100.00%	*BFSP2*	100.00%
BGN	100.00%	*BHLHA9*	100.00%	*BMP15*	100.00%
BPIFB6	100.00%	*CACNG2*	100.00%	*CAP1*	100.00%
CAPN10	100.00%	*CASP2*	100.00%	*CCDC8*	100.00%
CCNA2	100.00%	*CCNB3*	100.00%	*CD99*	100.00%
CDH3	100.00%	*CDK5R1*	100.00%	*CFAP47*	100.00%
CFP	100.00%	*CHM*	100.00%	*CHRDL1*	100.00%

CHUK	100.00%	CLCN5	100.00%	CMC4	100.00%
COL9A3	100.00%	*COMP*	100.00%	*CPD*	100.00%
CPXCR1	100.00%	*CRLF2*	100.00%	*CRYAA*	100.00%
CRYBA1	100.00%	*CRYBA4*	100.00%	*CRYBB1*	100.00%
CRYBB2	100.00%	*CRYBB3*	100.00%	*CRYGC*	100.00%
CRYGD	100.00%	*CSF2RA*	100.00%	*CSTF2*	100.00%
CTPS2	100.00%	*CXorf58*	100.00%	*CYP1B1*	100.00%
DACT1	100.00%	*DCHS2*	100.00%	*DDB2*	100.00%
DDX58	100.00%	*DECR1*	100.00%	*DGKH*	100.00%
DHRSX	100.00%	*DIAPH2*	100.00%	*DLL4*	100.00%
DMP1	100.00%	*DNAJC3*	100.00%	*DOCK11*	100.00%
DPF1	100.00%	*DPF3*	100.00%	*DSPP*	100.00%
EDA	100.00%	*EDNRA*	100.00%	*EIF2A*	100.00%
EIF2AK1	100.00%	*ELK1*	100.00%	*ENOX2*	100.00%
EOGT	100.00%	*EOMES*	100.00%	*ESX1*	100.00%
EVC	100.00%	*EVC2*	100.00%	*FAM111B*	100.00%
FAM161A	100.00%	*FAM20A*	100.00%	*FAM47B*	100.00%
FASN	100.00%	*FBP1*	100.00%	*FBXO25*	100.00%
FBXO8	100.00%	*FBXW4*	100.00%	*FEM1B*	100.00%
FKBP6	100.00%	*FKBPL*	100.00%	*FLT4*	100.00%
FOXC2	100.00%	*FOXE3*	100.00%	*FOXN1*	100.00%
FREM1	100.00%	*FYCO1*	100.00%	*FZD3*	100.00%
FZD6	100.00%	*GAB3*	100.00%	*GABRQ*	100.00%
GATA2	100.00%	*GDF5*	100.00%	*GJA3*	100.00%
GJA8	100.00%	*GLMN*	100.00%	*GON4L*	100.00%
GPR179	100.00%	*GPRASP1*	100.00%	*GRB14*	100.00%
GRM6	100.00%	*GSPT2*	100.00%	*GTPBP8*	100.00%
GUCY2C	100.00%	*HAUS7*	100.00%	*HDAC6*	100.00%
HNF4A	100.00%	*HOXA13*	100.00%	*HOXC13*	100.00%
HOXD13	100.00%	*HPGD*	100.00%	*HPSE2*	100.00%
HR	100.00%	*HSD3B7*	100.00%	*HSF4*	100.00%
IFITM5	100.00%	*IFNAR2*	100.00%	*IFT122*	100.00%
IFT80	100.00%	*IGSF1*	100.00%	*IHH*	100.00%
IL11RA	100.00%	*IL3RA*	100.00%	*INPPL1*	100.00%

INTS6L	100.00%	IQSEC3	100.00%	IRAK1	100.00%
KCNK12	100.00%	*KCNQ1*	100.00%	*KCTD1*	100.00%
KDM5A	100.00%	*KIF22*	100.00%	*KIF26B*	100.00%
KIT	100.00%	*KLF1*	100.00%	*KLF8*	100.00%
KLHL21	100.00%	*KLHL34*	100.00%	*KLHL4*	100.00%
LFNG	100.00%	*LHFPL3*	100.00%	*LIMK1*	100.00%
LOXHD1	100.00%	*LTBP2*	100.00%	*LTBP3*	100.00%
MACC1	100.00%	*MAGEA11*	100.00%	*MAGEB1*	100.00%
MAGEB10	100.00%	*MAGEB2*	100.00%	*MAGEC1*	100.00%
MAGEC3	100.00%	*MAGED1*	100.00%	*MAGEE2*	100.00%
MAGIX	100.00%	*MAP3K1*	100.00%	*MAP3K15*	100.00%
MAP3K7	100.00%	*MAP7D3*	100.00%	*MATN3*	100.00%
MBNL3	100.00%	*MC2R*	100.00%	*MCM9*	100.00%
MESP2	100.00%	*METAP1*	100.00%	*MGAT5B*	100.00%
MLH1	100.00%	*MMP21*	100.00%	*MORC4*	100.00%
MRAP	100.00%	*MTMR1*	100.00%	*MTMR8*	100.00%
MXRA5	100.00%	*MYH6*	100.00%	*MYH9*	100.00%
MYO1D	100.00%	*MYO1G*	100.00%	*MYT1*	100.00%
NCAPH	100.00%	*NECAB2*	100.00%	*NKX3-2*	100.00%
NOG	100.00%	*NPHS1*	100.00%	*NPHS2*	100.00%
NPR3	100.00%	*NR1I3*	100.00%	*NR2F2*	100.00%
NR5A1	100.00%	*NRK*	100.00%	*NSF*	100.00%
NTM	100.00%	*NXF4*	100.00%	*NXF5*	100.00%
OBSL1	100.00%	*ODF2L*	100.00%	*OR5M1*	100.00%
OTOGL	100.00%	*OTULIN*	100.00%	*P2RY4*	100.00%
P2RY8	100.00%	*PABPC5*	100.00%	*PAPSS2*	100.00%
PARP1	100.00%	*PASD1*	100.00%	*PAX9*	100.00%
PBRM1	100.00%	*PDE6G*	100.00%	*PECR*	100.00%
PGRMC1	100.00%	*PHF10*	100.00%	*PIK3C3*	100.00%
PIN4	100.00%	*PITX3*	100.00%	*PKD1L1*	100.00%
PKHD1	100.00%	*PLCE1*	100.00%	*PLCXD1*	100.00%
PMS2	100.00%	*POC1B*	100.00%	*POLD1*	100.00%
PPA2	100.00%	*PRDM12*	100.00%	*PRDX4*	100.00%
PRICKLE3	100.00%	*PRMT9*	100.00%	*PROX2*	100.00%

PRRG1	100.00%	PRRG3	100.00%	PRSS56	100.00%
RAPGEF1	100.00%	*RASA1*	100.00%	*RBL2*	100.00%
RBM28	100.00%	*RECQL4*	100.00%	*RENBP*	100.00%
RFX6	100.00%	*RGN*	100.00%	*RING1*	100.00%
RIPK4	100.00%	*RNPC3*	100.00%	*ROBO3*	100.00%
RRAS	100.00%	*RSPO4*	100.00%	*RTL9*	100.00%
RUNX2	100.00%	*RYR3*	100.00%	*SCARF2*	100.00%
SCRIB	100.00%	*SHROOM2*	100.00%	*SIX1*	100.00%
SIX5	100.00%	*SLC25A24*	100.00%	*SLC25A53*	100.00%
SLC25A6	100.00%	*SLC26A9*	100.00%	*SLC30A9*	100.00%
SLC31A1	100.00%	*SLC5A2*	100.00%	*SMARCA1*	100.00%
SMARCC1	100.00%	*SMARCD3*	100.00%	*SNTG1*	100.00%
SNX3	100.00%	*SOX17*	100.00%	*SPRTN*	100.00%
SPRY3	100.00%	*SREBF2*	100.00%	*SRY*	100.00%
STAB2	100.00%	*STARD8*	100.00%	*STAT5B*	100.00%
SYTL4	100.00%	*SYTL5*	100.00%	*TAB2*	100.00%
TAF7L	100.00%	*TBC1D8B*	100.00%	*TBX15*	100.00%
TBX20	100.00%	*TBX22*	100.00%	*TBX5*	100.00%
TBXAS1	100.00%	*TCEAL3*	100.00%	*TCP10L2*	100.00%
TEK	100.00%	*TENM1*	100.00%	*TEPSIN*	100.00%
TFB2M	100.00%	*TGFB2*	100.00%	*THUMPD1*	100.00%
TKTL1	100.00%	*TLR8*	100.00%	*TMEM132E*	100.00%
TMEM135	100.00%	*TMEM260*	100.00%	*TMPRSS6*	100.00%
TNKS2	100.00%	*TP73*	100.00%	*TPH2*	100.00%
TRAPPC2	100.00%	*TRAPPC6A*	100.00%	*TREX2*	100.00%
TRIM37	100.00%	*TSC22D3*	100.00%	*TSPAN8*	100.00%
TTC7A	100.00%	*TUBAL3*	100.00%	*TXNL4A*	100.00%
TYRP1	100.00%	*UROS*	100.00%	*USP18*	100.00%
UTP14A	100.00%	*UVSSA*	100.00%	*VAMP7*	100.00%
VIP	100.00%	*VSX2*	100.00%	*WDR13*	100.00%
WDR83OS	100.00%	*WNT10B*	100.00%	*WNT3*	100.00%
WNT4	100.00%	*WRN*	100.00%	*WT1*	100.00%
WWC3	100.00%	*XIAP*	100.00%	*XIST*	100.00%
XKRX	100.00%	*XPC*	100.00%	*ZBTB40*	100.00%

ZCCHC12	100.00%	ZCCHC8	100.00%	ZDHHC15	100.00%
PARP6	100.00%	*PIDD1*	100.00%	*RFX4*	100.00%
RFX7	100.00%	*MTCO2P12*	100.00%	*HCN2*	100.00%
RYR2	100.00%	*STARD7*	100.00%	*TXNRD1*	100.00%
CNTN2	100.00%	*CRH*	100.00%	*DMBX1*	100.00%
GAL	100.00%	*INO80*	100.00%	*MATN4*	100.00%
NID1	100.00%	*PCDHB4*	100.00%	*PRDM8*	100.00%
SCN2B	100.00%	*SLC7A6OS*	100.00%	*TUBA3E*	100.00%
FBXO28	100.00%	*RALGAPB*	100.00%	*SAMD12*	100.00%
CDK9	100.00%	*ECM1*	100.00%	*GOLGA2*	100.00%
PRIMA1	100.00%	*KIF26A*	100.00%		

Disclaimer:

1. The report is validated by third party lab
2. Test results are dependent on the quality of the sample received by the lab
3. Tests are performed as per schedule given in the test listing and in any unforeseen circumstances, report delivery may be delayed
4. Test results may show interlaboratory variations
5. All dispute and claims are subjected to local jurisdiction only. Clinical correlation advised
6. Test results are not valid for medico legal purposes
7. For all queries, feedback, suggestions, and complaints, please contact customer care support +0124 665 0000

End Of Report